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Enhancing the corrosion resistance of low-density steels via Al-rich oxide film formation by grain size optimization

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ABSTRACT

Fe-Mn-Al-C low-density steels have gained widely attention in recent years due to its low density and superior mechanical properties. This study investigated the effects of grain size on oxide film formation and corrosion behavior. Low-density steels with various grain sizes (8.7–32 μm) were prepared by adjusting the solution treatment temperature between 950 ℃ and 1150 ℃. The results indicate that as the solution treatment temperature increases, the grain size increases, and the grain boundary density decreases. The fine-grained sample (S950) rapidly forms an Al-rich oxide film. The film thickness reaches 25.2 ± 1.2 nm, with a low defect density ($0.68 \times 10^{21}/cm^3$ and $0.26 \times 10^{21}/cm^3$) and the lowest corrosion rate (0.160 mm/y). Conversely, the coarse-grained sample (S1150), characterized by a grain size of 32 μm, forms a thinner oxide film (18 ± 0.9 nm) with a reduced Al content (90.17 at% to 69.91 at%) and a higher defect density, and the corrosion rate increases by 43 %. Further analysis reveals that smaller grain sizes and high-density grain boundaries promote the preferential dissolution of elements such as Al at the grain boundaries. This drives the formation of a uniform and stable oxide film. The coarse-grained structure results in discontinuous oxide film growth due to sparse grain boundaries, which increases the Fe/Mn content, and diminishes the corrosion resistance of steels.

1. Introduction

Fe-Mn-Al-C low-density steels (LDSs) have emerged as promising materials for automotive and marine industry because of their lower density and outstanding mechanical properties[1–6]. These characteristics are mainly attributed to the high Mn and Al contents in LDSs [7–10], which effectively support the demand for lightweight and high-performance steel.

Moreover, prolonged service duration places greater demands on corrosion resistance, which imposes limitations on the long-term operational safety of LDSs. A higher Mn content in steel results in an increased proportion of weakly protective Mn oxides in the corrosion product film, which deteriorates the corrosion resistance of the steel [11–14]. In contrast, increasing the Al content in the LDSs reduces the corrosion current density by promoting the formation of a dense Al_2O_3 layer and an $Al(OH)_3$ film on the steel surface. This enhances the compactness of the film, and ultimately lowers the corrosion rate[11, 15–19]. Several studies have shown that microalloying is an effective way to improve the corrosion resistance of LDSs. The addition of elements such as Cr, Mo, and Ni promotes the enrichment of beneficial elements (e.g., Al, Cr, Mo) within the passive film, increases its thickness, and reduces defect density, thereby improving the stability and pitting resistance of the films. Table 1 summarizes the corresponding studies[12,13,20–25].

In summary, previous studies concerning the corrosion resistance of LDSs have primarily concentrated on enhancing oxide film density and stability through compositional optimization via alloying elements. On the other hand, microalloying increases the smelting cost and promotes the precipitate formation, thus posing a challenge to the uniformity of the structure. Consequently, an alternative strategy involving composition preservation coupled with microstructure optimization through heat treatment has emerged as a promising approach for corrosion

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Table 1
Summary of corrosion-related literature on LDSs.

Materials	Treatment	Phenomenon	Mechanism	Ref.
Fe-xMn−3Al−3Si	+ 21–28 % Mn	Reduced polarization resistance and increased corrosion current density	Excess Mn content leaded to insufficient Al enrichment in the oxide films	[12]
Fe−25Mn−10Al−1.5 C	+ 2 % Cr, + 2 % Ni	Increased OCP and polarization resistance, reduced corrosion rate	Suppressed early micro-galvanic effect, improved protective ability of passive film	[13]
Fe−30Mn−10.5Al−1.1 C	+ 3 % Cr, + 3 % Mo	Increased pitting potential	The passive film thickened and promoted the formation of Mo/Cr-rich passive film	[20]
Fe−30Mn−8Al−1.2 C	+ 2 % Cr, + 2 % Mo	Decreased corrosion current density	Formed a more protective oxide film and reduced amount of detrimental Mn oxides	[21]
Fe−19Mn−12Al−1.5 C	+ 5 % Cr	Increased pitting potential but reduced film thickness	Increased the Cr and Al content in the passive film and increased the oxide to hydroxide ratio	[22]
Fe−30Mn−8Al−1C	+ 3 % Cu	Increased carbide formation and accelerated corrosion rate	Cu enriched at the GBs and combined with dissolved oxygen to form an unstable oxide film	[23]
Fe−21Mn−10Al−1.25 C	+ 3 % Ni	Increased corrosion rate	B2 phase at GBs enhanced microgalvanic corrosion, leading to inhomogeneous film	[24]
Fe−20Mn−11.5Al−1.4 C	+ 5 % Cr	Higher pitting potential compared with stainless steel	Formed passive films thicker than stainless steel, with higher Cr+Al contents	[25]

resistance improvement. In recent years, some research has focused on the critical role of grain size in the corrosion process. Kim et al. [26] investigated the initial oxidation behavior of austenitic stainless steel and observed the formation of a double-layer oxide structure near grain boundaries, whereas nonprotective oxides predominantly formed within the interior of the grains. Grain refinement was shown to suppress the formation of rapidly growing spinel oxides, thereby enhancing oxidation resistance. Nnoka et al. [27] reported that grain refinement increased passivation coverage by approximately 40 %, with protective oxide layers preferentially forming along grain boundaries. Mohamed et al. [28] observed thicker oxide films at high-angle random GBs than at adjacent grains. This suggests greater anodic activity at the grain boundaries, which facilitates denser oxide formation. Li et al. [29] further demonstrated that an increased fraction of special grain boundaries in weathering bridge steel significantly improved corrosion resistance. Additionally, Shi et al. [30] found that during the initial stage of oxidation, GBs with relatively lower potentials typically act as anodic regions and undergo dissolution. An appropriate GB density helped homogenize the anodic oxidation current density and made the film growth more uniform.

Although these studies have established the influence of grain size on oxide film characteristics in alloys such as stainless steel, they have mainly focused on the electrochemical behavior or film thickness variations within conventional alloy systems. In the Fe-Mn-Al-C LDSs system, numerous studies[23,31–33] have investigated the corrosion behavior of alloys with comparable compositions after similar heat treatment (solution+aging). These studies predominantly attribute variations in corrosion resistance to precipitate characteristics. For example, the formation of precipitates such as carbides not only depletes the Al element in the matrix, but also intensifies the microgalvanic coupling between phases. This phenomenon promotes the formation of corrosion nucleation sites and accelerates the corrosion process. Furthermore, the segregation of precipitates at GBs induces preferential corrosion of GBs, which reduces the overall corrosion resistance. In addition, some studies[22] pointed out that κ-carbide and $D0_3$ precipitates frequently serve as the initiation sites for corrosion, reducing the pitting resistance of LDSs. The distribution and morphology of the precipitates significantly influence corrosion resistance. However, a comparison of these references[23,31–33] reveals that the reported grain sizes differ significantly (9.6 μm-50.2 μm), and the corresponding pitting potentials also vary considerably (-215 mV∼-480 mV) under similar heat treatment conditions. This phenomenon remains inadequately explained in current literature. Similar phenomena were also found in other LDS investigations[1,22,34,35]. Therefore, in addition to the effect of precipitates, the grain size evolution during solution treatment may constitute a critical factor governing the corrosion resistance of LDSs.

Therefore, in this study, the interference of precipitates was eliminated and Fe-Mn-Al-C LDSs with a gradient distribution of grain sizes (8.7–32 μm) were prepared via solution treatment, and the effects of the grain size and GB density on evolution of the Al-enrich oxide film were investigated. The microstructural characteristics of samples with varying grain sizes were systematically characterized by scanning electron microscopy (SEM) and electron backscatter diffraction (EBSD) techniques. The corrosion resistance of the LDSs was comprehensively assessed through electrochemical testing and immersion experiments. Furthermore, X-ray photoelectron spectroscopy (XPS) and Auger electron spectroscopy (AES) were employed to analyze the composition and structure of the oxide film. Finally, this study reveals the effect of grain size on the compositional distribution and formation behavior of Al-rich oxide films on LDSs. We found that grain refinement increased the Al content in the oxide film and improved the film uniformity. This discovery not only enriches the understanding of the surface oxidation behavior of LDSs, but also provides new ideas and theoretical support for improving the corrosion resistance of materials through microstructure regulation.

2. Experimental methods

2.1. Material preparation

The Fe-Mn-Al-C LDSs (including 30.34 % Mn, 8.93 % Al, and 0.9 % C) were prepared via a vacuum melting furnace. The steel ingot was homogenized at 1200 °C for 2 h, hot rolled at 1200–950 °C into slabs with a thickness of 20 mm, and quenched in water. The samples were subsequently subjected to solution treatment at 950 °C, 1050 °C, and 1150 °C for 2 h with water quenching to room temperature. For convenience, the original samples without solution treatment and the samples with different grain sizes obtained after solution treatment were labeled as S0, S950, S1050, and S1150, respectively.

2.2. Microstructural characterization

Samples with a size of 10 mm× 10 mm× 3 mm were sectioned from the heat-treated steel slab along the rolling direction. The surfaces were sequentially ground with SiC abrasive paper and mirror-polished with a diamond suspension. After being etched with 4 % nitric acid, the microstructure of the LDSs was analyzed by a JSM 7200 F SEM and matched energy dispersive spectroscopy (EDS). X-ray diffraction (XRD) and EBSD were used to investigate the crystal structure and phase composition of the LDSs. The scanning range of the XRD (using Cu Kα radiation) test was 30–90° with a scan rate of 4°/min. The samples for the EBSD test were electrolytically polished in a solution of 10 % CH_3COOH and 90 % C_2H_5OH, and EBSD data processing was performed

via AZtecCrystal software. The transmission electron microscopy (TEM) morphology and diffraction patterns of the LDSs were observed using FEI Tecnai G2 F20. The samples for TEM were prepared by double jet electropolishing using a mixture of 10 % perchloric acid and 90 % ethanol. The potential difference between the GB and the steel substrate was measured using a Bruker MultiMode8 Scanning Kelvin Probe Force Microscope (SKPFM) equipped with a PFQNE AL Kelvin probe.

2.3. Electrochemical measurements

For electrochemical testing, several samples with dimensions of 10 × 10 × 3 mm were prepared. The back of each sample was welded with copper wire and sealed with epoxy resin to leave a working area of 1 cm^2. After polishing with SiC sandpaper, the electrochemical measurements were performed on a Solartron Modulab XM electrochemical workstation. A three-electrode system was adopted, with a platinum sheet as the counter electrode, a saturated calomel electrode as the reference electrode, the prepared LDSs as the working electrode, and the corrosive medium was 3.5 wt% NaCl solution. To ensure consistency in the experimental conditions, pure N_2 was bubbled through the solution for 2 h to deoxygenate it before the experiment. The open-circuit potential (OCP) was first measured for 3600 s. The frequency range of electrochemical impedance spectroscopy (EIS) was from 10^5 Hz to 0.01 Hz with an excitation voltage of 10 mV. The potentiodynamic polarization curve was tested by scanning from 0.4 V below the OCP to the anodic current density of 5 mA/cm^2 with a scan rate of 0.3333 mV/s. The scanning potential range of the cyclic voltammetry (CV) test was −1.2 V_{SCE} to −0.3 V_{SCE}, and the scanning rate was 5 mV/s. All the electrochemical tests were repeated at least three times, and the error was controlled within 5 %. The consistency of the data demonstrated the good repeatability of the experiment.

2.4. Immersion test

The corrosion resistance of the LDSs was evaluated through immersion testing in 3.5 wt% NaCl solution. The size of samples was 30 × 25 × 3 mm. The solution was replaced every 7 days, and the immersion period was 30 days. The corrosion rate was calculated by Eq. (1).

$$v = \frac{(W_0 - W_t) \times 8.76 \times 10^4}{\rho S t} \quad (1)$$

where v represents the corrosion rate, mm/y; W_0 is the initial weight of the sample, g; W_t is the weight of the sample after rust removal, g; S is the immersion area, cm^2; t is the immersion time, d; ρ is the density of the LDSs, g/cm^3.

2.5. Oxide film analysis

The composition of the oxide film on the LDSs surface was characterized using XPS and AES. Prior to the constant potential polarization test, the samples were cathodically polarized by applying −1.0 V_{SCE} for 300 s to remove the air-formed oxide film, and then the samples were polarized at −0.5 V_{SCE} for 7200 s to form the oxide film on the surface. After polarization, the samples were immediately rinsed with deionized water, dried with anhydrous ethanol, and then vacuum-sealed to prevent surface oxidation. XPS was conducted using a ThermoFisher ESCALAB 250Xi XPS instrument with an Al Kα anode target at a power of 72 W. The spectra were corrected using the standard carbon peak (284.8 eV) and analysed using Avantage software. The AES measurements were performed with a ULVAC-PHI PHI-700 nanometer scanning Auger probe operating at an accelerating voltage of 5 kV. The sputtering rate was 3.6 nm/min (relative to the thermally oxidized SiO_2 film on the surface of single crystal Si), and the collection area for the Auger electron signal was 70 × 70 μm. Mott-Schottky (M-S) analysis was used to investigate the semiconductor properties of the oxide film on the LDSs surface. The testing potential ranged from −0.5 V_{SCE} to −1.1 V_{SCE}, and the scan step was 30 mV/s.

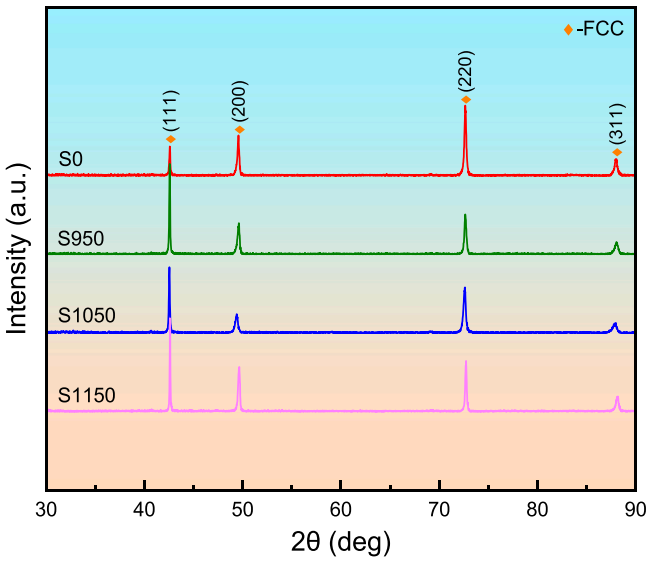


Fig. 1. XRD spectra of the LDSs after solution treatment.

3. Results

3.1. Microstructure analysis

The main phase composition of the LDSs was analyzed by XRD, and the diffraction patterns are shown in Fig. 1. The diffraction peaks reveal a single-phase face-centered cubic (FCC) austenitic structure, as confirmed by the presence of (111), (200), (220) and (311) peaks. These results indicate that the solution treatment preserves the austenitic matrix while modifying the relative peak intensities. Notably, the (111) peak intensity significantly increases following solution treatment, whereas the (220) peak intensity correspondingly decreases.

Fig. 2 presents the microstructure images of the LDSs after different solution temperatures. The microstructures of all the samples exhibit a single-phase austenitic structure. The grain size of the S0 sample without solution treatment is uneven, featuring both large deformed grains and fragmented small grains resulting from thermal deformation during rolling (Fig. 2a). With increasing solution treatment temperature, the grain size gradually increases, as shown in Fig. 2b-d.

The crystallographic characteristics of the LDSs were investigated through EBSD, and the results are shown in Fig. 3, and Fig. 4 summarizes the average grain size, kernel average misorientation (KAM) value, and GB density for all samples. The inverse pole figure (IPF) maps in Fig. 3b-d reveal that LDSs do not exhibit significant preferred orientations after solution treatment and that the crystal structure orientations have a random crystallographic orientation distribution. Fig. 3a indicates that the S0 sample has a nonuniform grain size distribution, and some small grains are distributed at the interface of large grains. Solution treatment effectively homogenizes the grain size, as evidenced in Fig. 3b-d, and the grain size gradually increases with temperature. The maximum average grain size of 32.0 μm is achieved at the highest solution temperature of 1150 ℃ (Fig. 4a-d).

The KAM mapping was employed to qualitatively evaluate the distribution of local residual stresses within the materials, where elevated KAM values correspond to increased residual stress accumulation and consequently higher corrosion susceptibility[36]. Fig. 3a_1-d_1 present the KAM distributions for the samples. The S0 sample exhibits the most pronounced residual stress, with an average KAM value of 1.62 as shown in Fig. 4e. After solution treatment, the sub-grain boundaries undergo

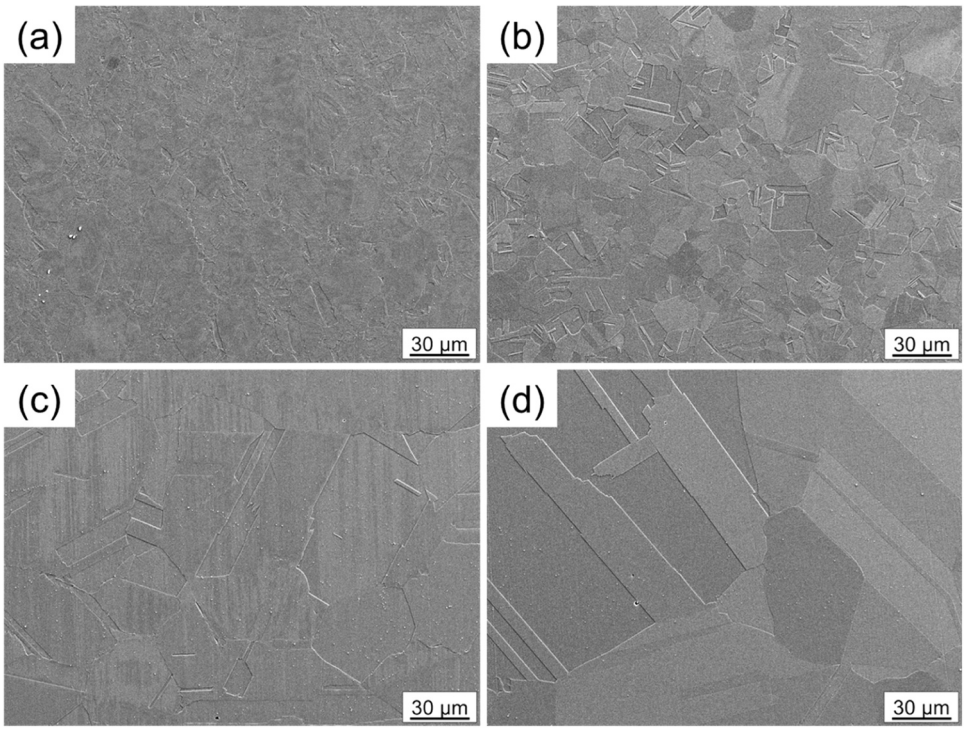


Fig. 2. Microstructure of the LDSs after solution treatment: (a) S0, (b) S950, (c) S1050, (d) S1150.

dynamic recrystallization, reducing the residual stress in the S950, S1050, and S1150 samples to an average KAM value of approximately 0.4. The GB distribution analysis in Fig. 3a_2-d_2 reveals that the S0 sample contains a substantial population of low-angle grain boundaries (LAGBs, 2–10°). After solution treatment, the S950, S1050, and S1150 samples exhibit complete recrystallization[37], characterized by abundant annealing twin boundaries (ATBs, Σ3, 60°) and high-angle grain boundaries (HAGBs, >10°) within equiaxed austenite grains. Fig. 4e indicates that the S950 sample has the maximum densities of both ATBs and HAGBs, which are 0.146 and 0.114 μm/μm^2, respectively. These boundary densities progressively decrease with increasing solution temperature, with the S1150 sample reaching minimum values of 0.040 and 0.019 μm/μm^2, respectively.

TEM characterization was conducted to investigate the microstructural characteristics of the LDSs, and the results are shown in Fig. 5. The bright-field TEM image in Fig. 5a, along with the corresponding high-resolution TEM (HRTEM) image and fast Fourier transform (FFT) analysis (Fig. 5a_1), confirms the presence of numerous κ-carbides (indicated by yellow arrows) with diameters ranging from 5 to 10 nm in the S0 sample. Fig. 5a_2 reveals that the matrix of the LDSs is predominantly austenitic, and contains a high density of irregularly distributed dislocations in the S0 sample. As shown in Fig. 5b, the dislocation density decreases after solution treatment. Notably, dislocation tangles are observed adjacent to the ATBs, suggesting that the abundant ATBs effectively hinder dislocation movement.

3.2. Electrochemical properties

Fig. 6 presents the potentiodynamic polarization curves and corresponding corrosion current densities of the LDSs in 3.5 wt% NaCl solution. As shown in Fig. 6a, the cathodic polarization curves exhibit nearly identical behavior across all the samples nearly overlap, primarily involving the oxygen absorption reaction. This indicates that the change in grain size caused by solution treatment has a negligible influence on the cathodic reaction. The anodic polarization process reveals three distinct regions. In the potential range of −0.91 V_{SCE} to −0.67 V_{SCE}, the current density rises with increasing potential, which corresponds to the active dissolution region. From −0.67 V_{SCE} to −0.58 V_{SCE}, the current density decreases, and the electrode reaction is in an unstable state, which indicates the surface oxide film formation. Subsequently, with the increase of potential, the current density remains relatively constant. This may be related to the formation of an oxide film with certain protective capabilities, and a similar passive phenomenon occurs. A breakdown current density (i_{bre}) derived from the polarization curves is shown in Fig. 6b. The lower breakdown current density indicates better corrosion resistance[38,39]. Notably, the error bars in this figure represent the standard deviation range of the tests. The error values of all the samples are controlled within 5 %, and the experimental data have good repeatability and reliability. The S0 sample without solution treatment has the highest i_{bre} (9.92 μA/cm^2), and is relatively more prone to corrosion. The i_{bre} of the fine-grained sample (S950) is 2.72 μA/cm^2, which is lower than that of the coarse-grained sample (S1150) at 8.84 μA/cm^2, and it has higher corrosion resistance.

The surface corrosion morphology of the LDSs after potentiodynamic polarization to 5 mA/cm^2 is shown in Fig. 7. According to Fig. 7a, the S0 sample exhibits extensive corrosion damage, characterized by numerous deep pits resulting from elevated surface residual stress and a heterogeneous GB distribution (Fig. 3). After solution treatment at 950 °C, the LDS shows slight pitting corrosion (Fig. 7b), with smaller pit sizes, indicating superior corrosion resistance. However, as the grain size increases (Fig. 7c-d), the number and size of corrosion pits in the S1050 and S1150 samples increase, and corrosion expansion occurs on some pit walls, and the corrosion resistance decreases.

Fig. 8 shows the EIS results of the LDSs in 3.5 wt% NaCl solution. The Nyquist plots in Fig. 8a reveals that all the samples display a single capacitive arc without Warburg impedance. This behavior suggests that the electrode reactions are predominantly controlled by electrochemical reactions. The arc radius of the capacitance represents the resistance to corrosion, where a larger capacitive arc radius corresponds to enhanced corrosion resistance[40]. The S0 sample without solution treatment displays the smallest capacitive arc radius, which demonstrates its greater susceptibility to corrosion. In contrast, the S950 sample has the largest capacitive arc radius, which indicates that it has the highest resistance and the lowest corrosion tendency.

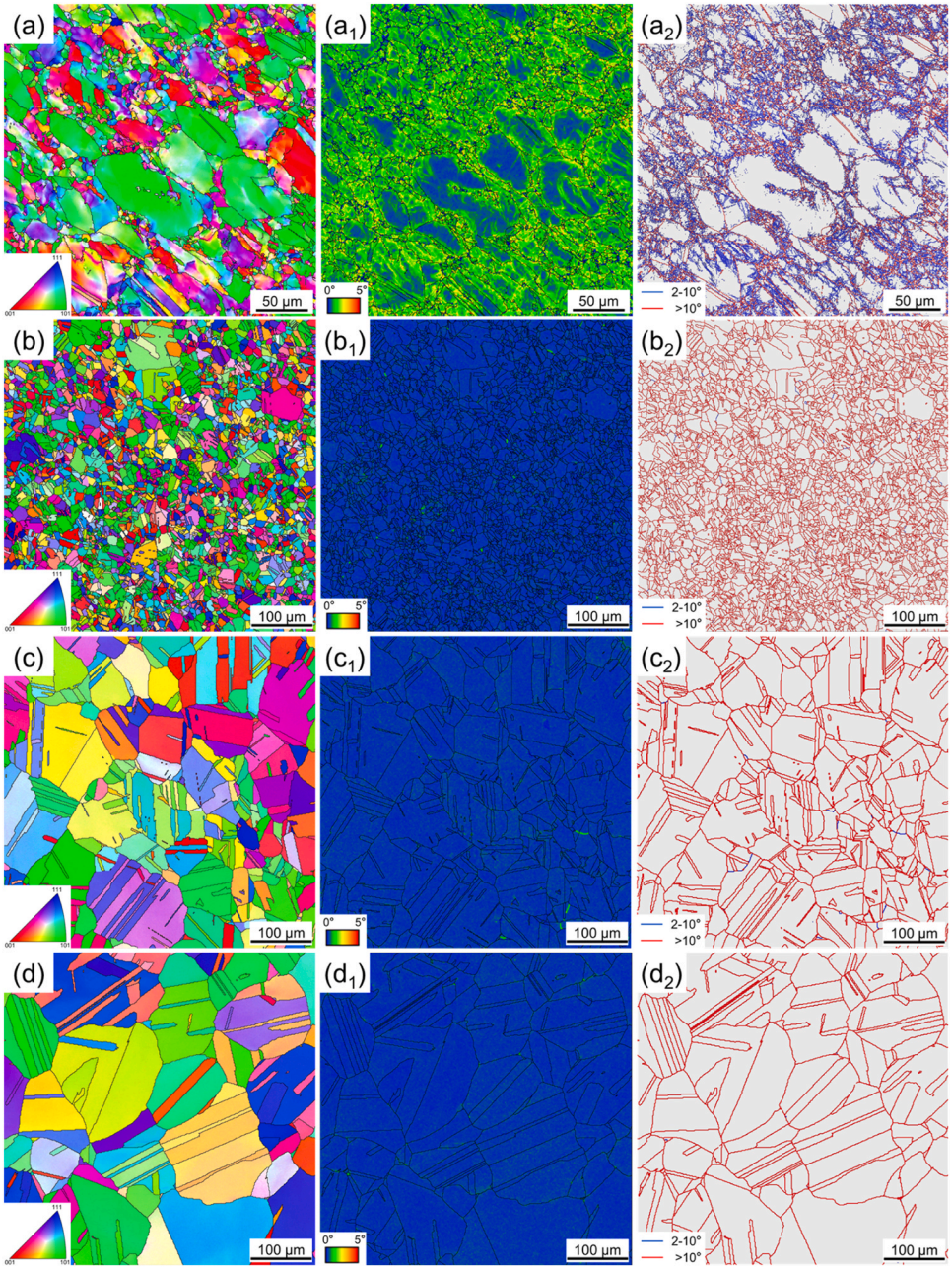


Fig. 3. EBSD images of the LDSs: (a-a_2), (b-b_2), (c-c_2), and (d-d_2) are IPF, KAM, and GB distribution maps of S0, S950, S1050, and S1150 samples, respectively.

The EIS data of the LDSs were fitted via an appropriate equivalent circuit model. According to the Bode phase plot in Fig. 8c, the EIS results exhibit the superposition of two times constants, with a maximum phase angle between 65° and 75° at intermediate frequencies, characteristic of typical nonideal capacitive responses[41,42]. This deviation from the ideal capacitance arises from electrode surface porosity, inhomogeneity, and relaxation effects[43]. Consequently, the constant phase angle element Q is used in the equivalent circuit instead of the ideal capacitance C to account for these nonideal characteristics[44,45]. On the basis of the above analysis, the equivalent circuit diagram in Fig. 8a is employed for EIS fitting, where R_s is the solution resistance, R_1 and Q_1 are the diffusion resistance and constant phase angle element of ions at the solution/film interface, respectively, and R_2 and Q_2 are the oxide film resistance and constant phase angle element, respectively. The impedance of the two constant phase elements (CPEs), Q_1 and Q_2, in the equivalent circuit is defined as[46,47]:

$$Z_{\mathrm{CPE}} = \frac{1}{Q(j\omega)^{\mathrm{n}}} \tag{2}$$

where Q is the admittance of the CPE, j is the imaginary unit, ω is the angular frequency, and n is the dispersion index. When $0.5 < \mathrm{n} < 1$, the CPE represents a nonideal capacitor, and when $\mathrm{n} = 1$, the CPE represents an ideal capacitor.

Fig. 8a-c demonstrate excellent agreement between the experimental data and fitted curves. This finding indicates that the equivalent circuit is appropriate and accurately reflects the electrochemical behavior of the system. The polarization resistance R_p (R_p=R_1+R_2) is used to evaluate the corrosion resistance of materials[48]. The fitting results are shown in Table 2. Comparative analysis reveals that the S950 sample exhibits the highest R_p (12486 Ω·cm^2), which is approximately double that of the S0 sample (6036 Ω·cm^2). Additionally, the R_p values progressively decrease with increasing solution temperature. This trend is

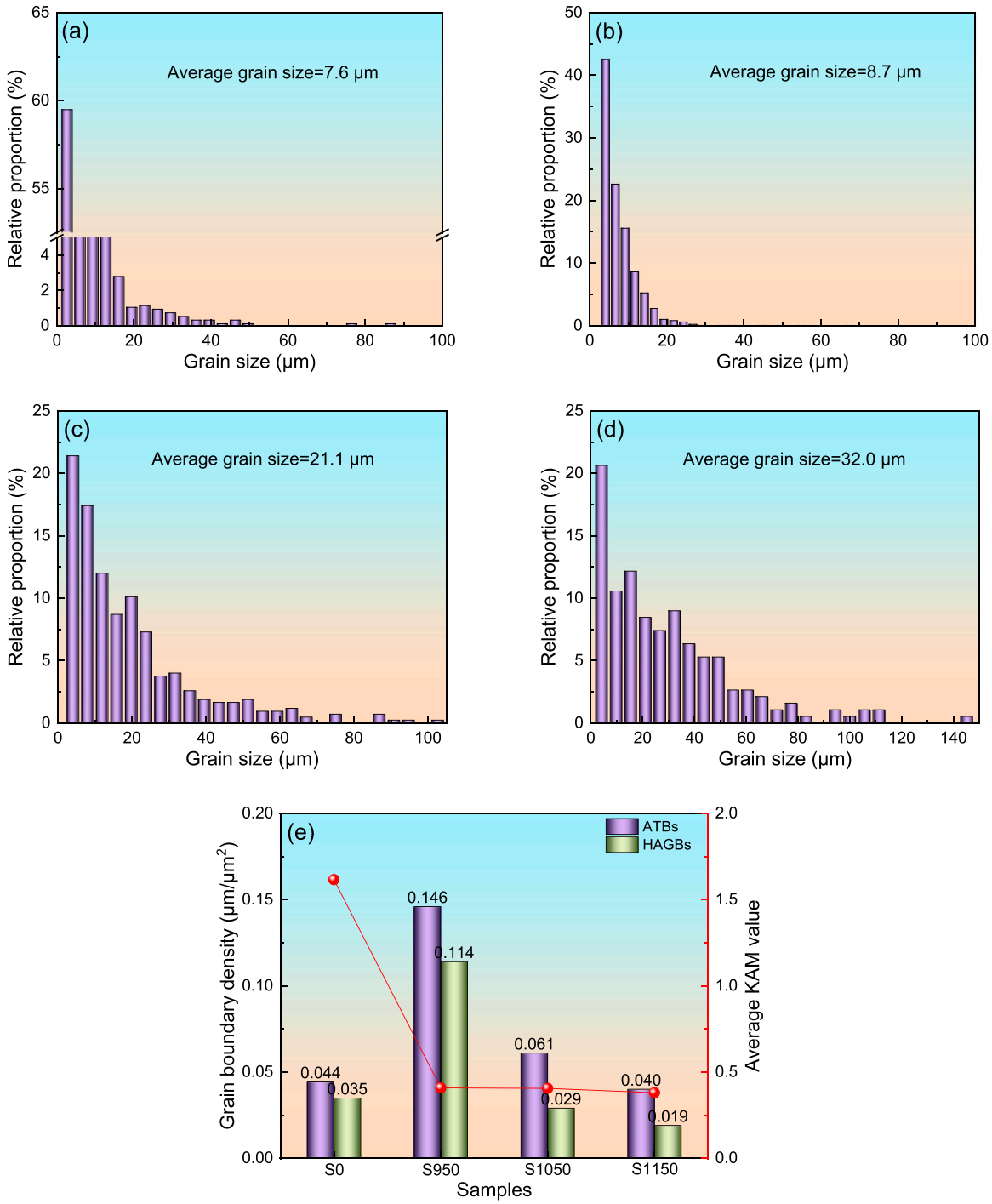


Fig. 4. The grain size distribution of the LDSs: (a) S0, (b) S950, (c) S1050, (d) S1150, and (e) the result of KAM value and the GB density.

further supported by the Bode magnitude plots in Fig. 8b. The magnitude corresponding to 0.01 Hz can be used to evaluate the corrosion resistance of materials. Higher values correspond to better corrosion resistance[44]. The S950 sample clearly has the highest magnitude at this frequency, whereas the S0 sample exhibits the lowest magnitude. These findings are consistent with the results obtained from the potentiodynamic polarization results, collectively demonstrating the superior corrosion resistance of the S950 sample.

Fig. 9 presents the corrosion rate of the LDSs after 30 days of immersion in 3.5 wt% NaCl solution. The S0 sample has the highest corrosion rate (0.326 mm/y), reflecting its inferior corrosion resistance. Solution treatment significantly reduces the corrosion rate, and the S950 sample exhibits the lowest corrosion rate at 0.160 mm/y. This suggests that grain refinement improves the corrosion resistance of steel. As the grain size increases, the corrosion rate correspondingly increases. The corrosion rates of the S1050 and S1150 samples are 0.248 mm/y and 0.281 mm/y, respectively. In addition, the experimental data show excellent reproducibility, with variations maintained below 3 % across multiple tests. This also further proves that the effect of grain size on the structure of the oxide film is consistent among the different samples. In summary, the difference in grain size caused by solution treatment has a great influence on the corrosion behavior of LDSs, and the optimal corrosion resistance is achieved at 950 °C. However, as the temperature increases, the corrosion resistance gradually decreases, which is

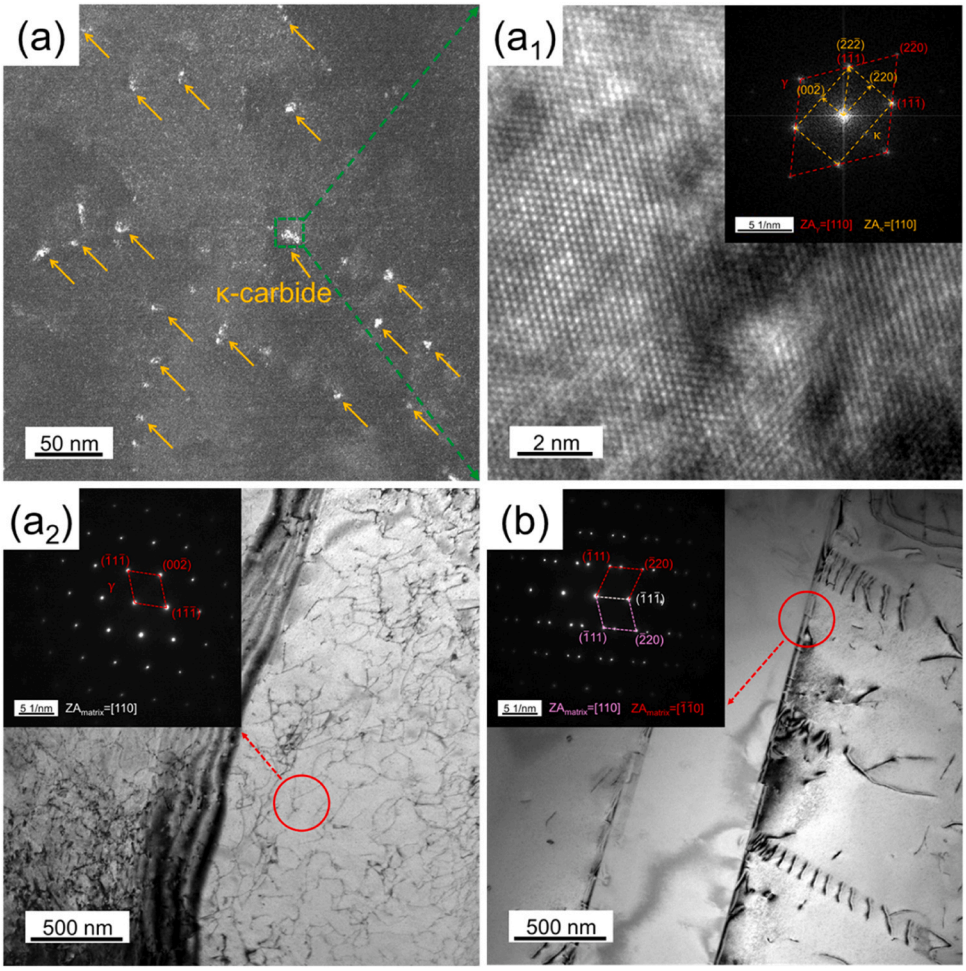


Fig. 5. TEM morphology of the LDSs: (a-a$_1$) TEM morphology and HRTEM image of κ-carbide in S0 sample; (a$_2$) dislocation morphology of S0; (b) twin morphology after heat treatment.

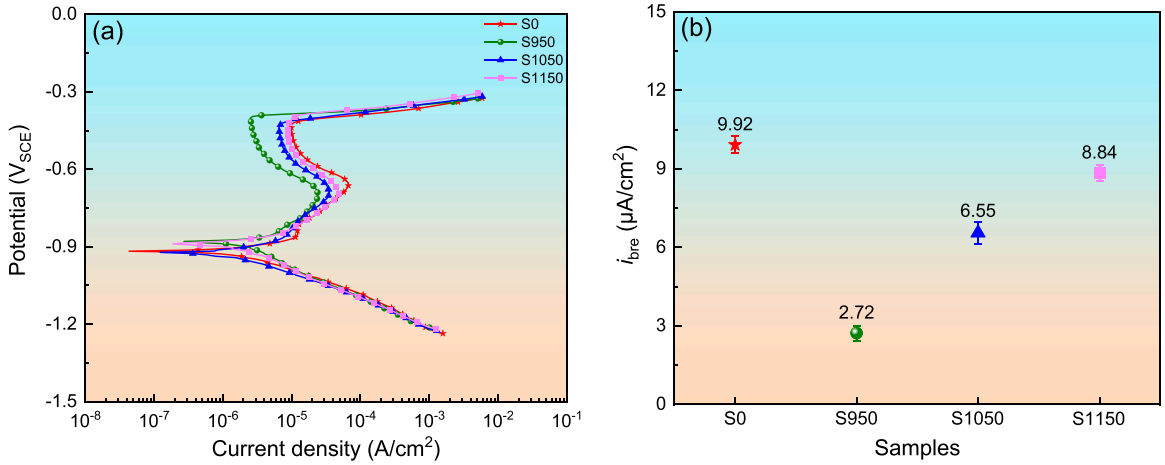


Fig. 6. (a) Potentiodynamic polarization curves and (b) the corresponding breakdown current density of LDSs in 3.5 wt% NaCl solution.

consistent with the results shown in Figs. 6 and 8, and may be related to the difference in grain size and the Al content in the oxide film.

3.3. Oxide film analysis

The corrosion resistance of the LDSs was further evaluated through potentiostatic polarization tests, and the relationship between the current density and time is presented in Fig. 10. During the initial 60 s of polarization, all samples exhibited a rapid current density decline to below 9 μA/cm^2, corresponding to surface oxide film formation and initial anodic dissolution. This behavior confirms that all the samples can quickly form oxide films to inhibit anodic dissolution within a short period[49,50]. With increasing polarization time, the current density of all samples increases and then decreases during the polarization period

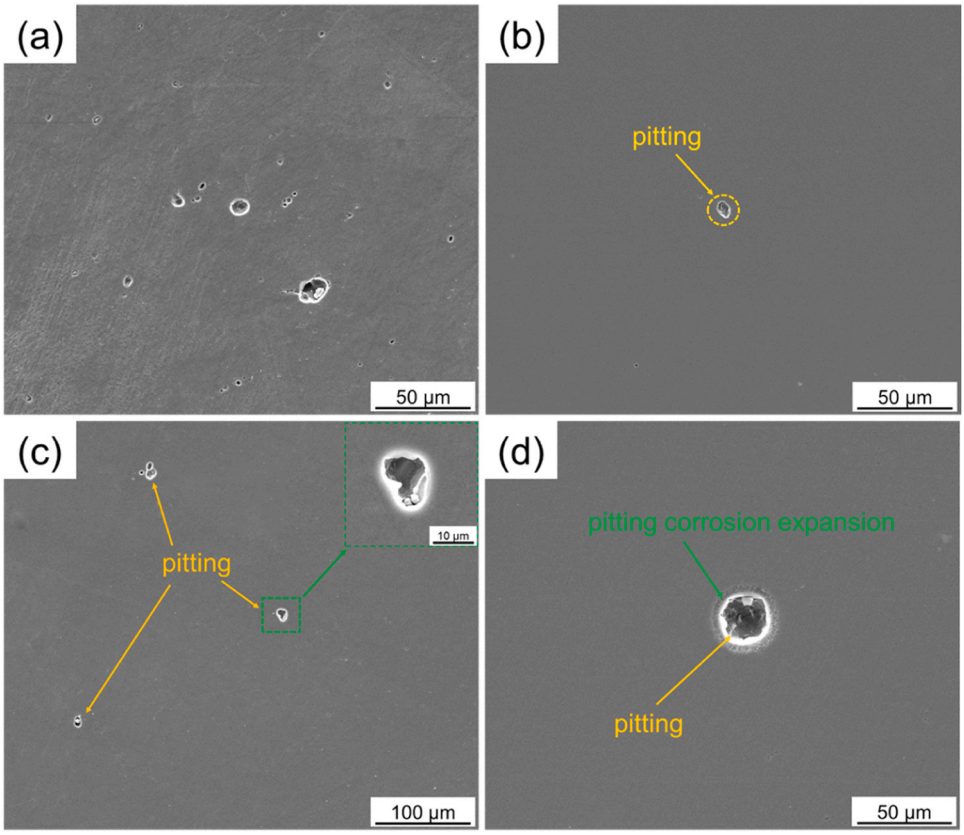


Fig. 7. Surface corrosion morphology of the LDSs after potentiodynamic polarization: (a) S0, (b) S950, (c) S1050, (d) S1150.

of 600–1400 s, which corresponds to the rupture and self-healing process of the oxide film[51,52]. The S0 sample exhibits the highest stable current density (8.39 μA/cm^2), which reflects relatively weak protection of its oxide film. With the increase of solution temperature, the grain size increases, and the stable current density of the LDSs gradually increases. The S950 sample achieves the lowest stable current density (2.45 μA/cm^2), outperforming both S1050 (4.52 μA/cm^2) and S1150 (6.04 μA/cm^2). These results demonstrate that the oxide film of the S950 sample provides optimal corrosion protection.

The EIS characterization of the oxide film on the LDSs after potentiostatic polarization for 7200 s in 3.5 wt% NaCl solution is presented in Fig. 11. The Nyquist plot in Fig. 11a reveals that the S0 sample has the smallest capacitive arc radius, indicating poor protective performance of its oxide film and higher corrosion susceptibility. The capacitance arc radius decreases with grain growth. The S950 sample has the largest radius, which confirms the superior density and stability of its oxide film. The Bode modulus plot in Fig. 11b indicates that the impedance modulus value at low frequency decreases with increasing grain size. The S950 sample maintains the highest low-frequency impedance, confirming the strong protective capability of its oxide film. The Bode phase plot (Fig. 11c) also shows a similar pattern. The S950 sample maintains a higher phase angle over a wider frequency range. These features demonstrate that its oxide film structure is more stable and uniform. The equivalent circuit diagram of Fig. 11a was used for EIS fitting, and the fitting results are shown in Table 3. The S950 sample achieves the highest R_p of 21355 Ω·cm^2. Its oxide film has superior uniformity and density, which provide optimal protective properties.

To quantitatively assess the corrosion protection ability of the oxide films of samples with different grain sizes, the protection efficiency (*PE*) of the oxide film was calculated from the R_2 values via Eq. (3)[53,54]:

$$PE = \left(1 - \frac{R_2^0}{R_2^c}\right) \times 100\% \tag{3}$$

where R_2^0 is the interfacial resistance before film formation, and R_2^c is the resistance of the oxide film after film formation. The calculation results show that the fine-grained sample (S950) has a greater *PE* (30.75 %) compared to the coarse-grained S1150 sample (17.95 %), confirming that grain refinement enhances the protective properties of oxide film. This improvement can be attributed to the high GB density facilitating the formation of an Al-rich oxide film. Notably, although the *PE* has improved, the overall *PE* values remain relatively low, likely due to insufficient oxide film compactness and localized structural defects.

The oxide film formed on the surface of LDSs exhibits semiconductor characteristics. Oxide films with different compositions and structures have varying carrier concentrations (defect densities), which directly influence the corrosion resistance. The semiconductor behavior of the oxide film can be quantitatively evaluated through space charge layer capacitance measurements and the relationship curve of the electrode potential (Mott-Schottky curve), with the equation as follows[55,56]:

$$\frac{1}{C_H^2} + \frac{1}{C_{SC}^2} = \pm\frac{2}{\varepsilon\varepsilon_0 eN}\left(E - E_{fb} - \frac{kT}{e}\right) \tag{4}$$

where C_H is the Helmholtz double layer capacitance, C_{SC} is the space charge layer of the oxide film, "+ " is the n-type semiconductor, "-" is the p-type semiconductor, e is the electron charge (1.6 ×10^{-19} C), ε is the relative dielectric constant of the semiconductor, which is 15.6, ε_0 is the dielectric constant of vacuum, which is 8.85 × 10^{-12} F/m, N is the carrier density (where N_D is the donor density corresponding to the n-type semiconductor and N_A is the acceptor density corresponding to the p-type semiconductor), k is the Boltzmann constant, T is the absolute temperature, E is the applied potential, and E_{fb} is the flat band potential. Given the significantly lower magnitude of space charge capacitance than that of the Helmholtz capacitance, Eq. (4) can be simplified to Eqs. (5) and (6).

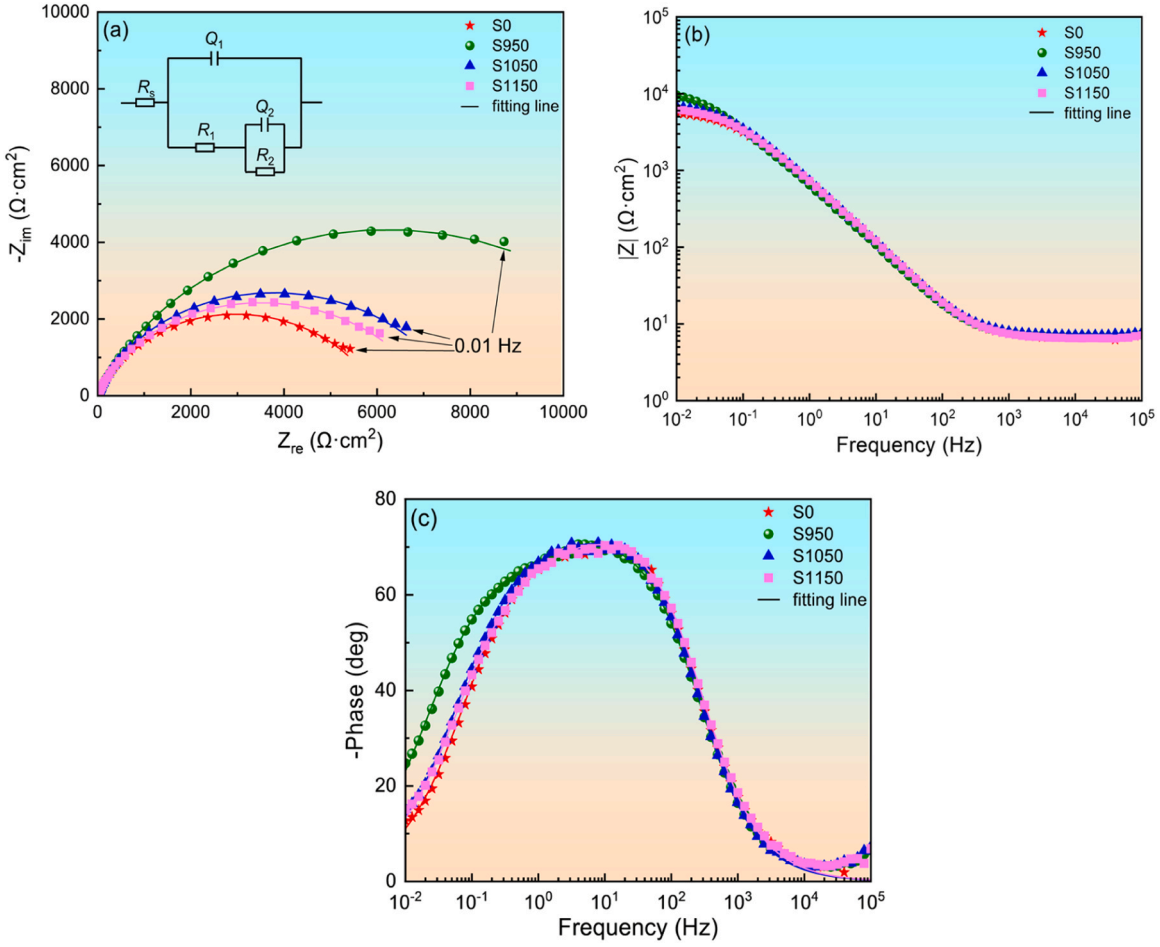


Fig. 8. EIS curves of the LDSs in 3.5 wt% NaCl solution: (a) Nyquist plot, (b) Bode magnitude plot, (c) Bode phase plot.

Table 2
Fitting parameters of the EIS curves of the LDSs.

	R_s (Ω·cm²)	Q_1 (Ω⁻¹·cm⁻²sⁿ)	n_1	R_1 (Ω·cm²)	Q_2 (Ω⁻¹·cm⁻²sⁿ)	n_2	R_2 (Ω·cm²)	R_p (Ω·cm²)
S0	6.55	2.62×10^{-4}	0.85	1216	1.13×10^{-4}	0.60	4820	6036
S950	6.80	2.92×10^{-4}	0.84	2092	1.43×10^{-4}	0.66	10394	12486
S1050	6.52	2.40×10^{-4}	0.86	1261	1.07×10^{-4}	0.59	6753	8014
S1150	6.87	2.42×10^{-4}	0.85	1036	1.36×10^{-4}	0.58	5956	6992

$$\frac{1}{C_{SC}^2} = \frac{2}{\varepsilon\varepsilon_0 e N_D}\left(E - E_{fb} - \frac{kT}{e}\right) (n-\text{type semiconductor}) \quad (5)$$

$$\frac{1}{C_{SC}^2} = \frac{-2}{\varepsilon\varepsilon_0 e N_A}\left(E - E_{fb} - \frac{kT}{e}\right) (p-\text{type semiconductor}) \quad (6)$$

According to the Mott-Schottky formula and the point defect model (PDM)[57,58] of the oxide film, the $1/C_{SC}^2$ is linearly related to E. A positive slope in the Mott-Schottky curve indicates n-type semiconductor behavior, characterized by cation interstitials and oxygen vacancies as the predominant defects. Conversely, a negative slope signifies p-type semiconductor properties, with cation vacancies dominating the defect structure[59,60]. The greater the donor or acceptor density of the oxide film on the surface of the LDSs, the more susceptible the oxide film is to damage during the corrosion process.

The Mott-Schottky analysis presented in Fig. 12 reveals the semiconductor characteristics and defect properties of oxide films formed on LDSs. As shown in Fig. 12a, all the samples demonstrate similar duplex semiconductor characteristics, exhibiting both n-type and p-type characteristics, across different potential regions. In Region 1 (-1.13 V_{SCE} to −0.93 V_{SCE}), the positive slope confirms that the n-type behavior is dominated by cation interstitials and oxygen vacancies. The slope of the curve in Region 2 (-0.93 V_{SCE} to −0.67 V_{SCE}) shows negative slope characteristics indicative of p-type behavior, with cation vacancies as the primary defects. The n-type semiconductor characteristic of LDSs is mainly controlled by substances such as Al_2O_3 and Fe_2O_3 in the film[61,62], whereas the p-type semiconductor characteristic is mainly controlled by oxides such as FeO[63,64]. The breakdown of the oxide film occurs at surface sites with high rates of cation vacancy diffusion and oxygen vacancy migration, where aggressive ions readily occupy these defect sites[65]. The carrier concentration in the oxide film of the LDSs is calculated by Eq. (4) and (5), and the N_D and N_A are shown in Fig. 12b. A lower N_D and N_A of the oxide film correspond to fewer structural defects and consequently reduced pathways for chloride ion penetration. The S0 sample exhibits the highest N_D (0.7×10^{21} /cm³) and N_A (2.25×10^{21} /cm³) values. This defective oxide film structure leads to inferior corrosion resistance. Although the solution treatment generally increases the N_D and N_A with increasing grain size, all the treated samples maintain lower values than the S0 sample. The S950 sample achieves the lowest N_D (0.26×10^{21} /cm³) and N_A (0.68×10^{21}

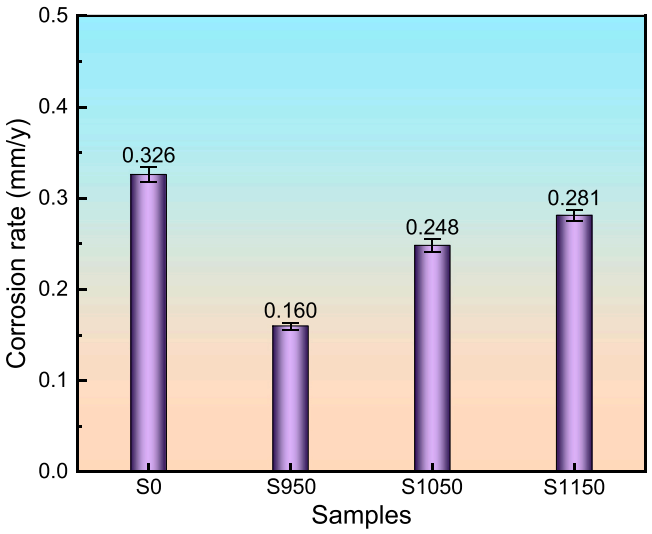


Fig. 9. Corrosion rate of the LDSs after immersion for 30 d in 3.5 wt% NaCl solution.

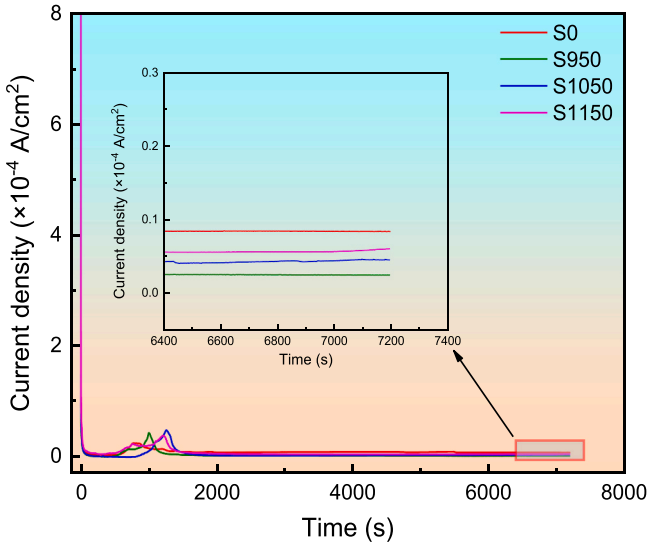


Fig. 10. Potentiostatic polarization curves of the LDSs in 3.5 wt% NaCl solution.

/cm^3) values, which is consistent with its superior oxide film compactness and corrosion resistance.

XPS analysis was performed to characterize the elemental composition and valence state distribution of the surface oxide film of the LDSs after 7200 s of potentiostatic polarization in a 3.5 wt% NaCl solution. Fig. 13 shows the XPS spectra of Fe2p3/2, Mn2p3/2, Al2p, and O1s in the oxide film of the samples. According to the Fe2p3/2 spectra of the oxide film (Fig. 13a-d), Fe is composed of Fe^{2+}_{OX} (709.8 eV), Fe^{3+}_{OX} (711.5 eV) and Fe^{3+}_{hy} (713.3 eV)[66–68]. Moreover, the S950 sample additionally exhibits a satellite peak Fe^{2+}_{sat} (715.0 eV), corresponding to the oxides FeO and Fe_2O_3 and the hydroxide FeOOH. Fig. 13a_1-d_1 shows the XPS spectra of Mn2p3/2 in the oxide film, where three peaks are detected in all the samples except for the S950 sample, composing of Mn^{2+} (640.4 eV), Mn^{4+} (641.8 eV), and the satellite peak Mn^{2+}_{sat} (646.2 eV)[69–71], corresponding to $Mn(OH)_2$ and MnO_2. The S950 sample uniquely displays only a Mn^{2+} (640.4 eV) oxidation state. The Al2p spectra (Fig. 13a_2-d_2) exhibit a single peak corresponding to Al^{3+} (74.1 eV)[22,72], which is attributed to Al oxide (Al_2O_3) and hydroxide ($Al(OH)_3$). According to the O1s spectra (Fig. 13a_3-d_3), the O in the surface oxide film consists of OH^- (531.6 eV), H_2O (533.2 eV), and O^{2-} (529.7 eV)[67,68,70,71,73], corresponding to hydroxides, crystalline water, and oxides, respectively.

The relative contents of different valence states of each metal in the oxide film of all the samples were calculated using Eq. (7), and are presented in Fig. 14.

$$C_X = \frac{I_X/S_X}{\sum I_i/S_i} \tag{7}$$

where I_X is the peak area corresponding to the element, and S_X represents the sensitivity factor of the XPS test. As shown in Fig. 14, Al emerges as the predominant metallic element in the oxide films of all the samples. Comparative analysis reveals that the S0 sample contains substantially higher Fe and Mn contents alongside a reduced Al content. The Al content progressively decreases with grain coarsening, accompanied by corresponding increases in the Fe and Mn contents, but the Al content in the samples after solution treatment is still higher than that in the S0 sample. The S950 sample has the highest content of Al in the oxide film, which is 90.17 at%. This behavior strongly indicates that grain refinement promotes the preferential dissolution and film formation of Al.

AES analysis was used to characterize the elemental distribution within the oxide film along the depth profile. Fig. 15 illustrates the atomic concentration profiles of O, Fe, Al, and Mn at various depths, which reveal oxygen enrichment at the surface of all samples with a gradual transition to the substrate composition at greater depths. The main components of the oxide film consist of Al and Fe oxides, and also contain trace amounts of Mn oxide. The oxide film thickness is conventionally determined as the depth at which the oxygen concentration decreases to half of its maximum value[74,75]. On the basis of this definition, the oxide film thickness of S0 is approximately 16.2 ± 0.8 nm and consists mainly of Fe, Al, and O, with Fe and Al present in roughly comparable amounts. In comparison, the S950 sample has a significantly thicker film of 25.2 ± 1.2 nm, which is predominantly composed of uniformly distributed Al oxides. The Fe and Mn contents in the film are relatively low, which suggests a more compact and stable oxide film. The thicknesses of the S1050 and S1150 oxide films are 21.6 ± 1 nm and 18 ± 0.9 nm, respectively. Although the Al content is high in the outer film, with increasing sputtering depth, the Fe content increases and the Al content decreases. In summary, both the oxide film thickness and Al content tend to decrease with increasing grain size. Notably, compared with the other samples, the S950 sample forms a uniform, dense, and thick Al-rich oxide film.

4. Discussion

4.1. Effect of solution temperature on the microstructure of LDSs

Figs. 2–4 indicate that after solution treatment, the grain size and GB density of the LDSs significantly change, whereas the phase composition and precipitates remain essentially unchanged. The driving force for grain growth originates from the reduction of the system's free energy following GB migration, which constitutes the primary mechanism for grain growth. The migration rate is predominantly governed by GB energy and atomic diffusion capability[76–78]. Based on the classic grain growth model, the evolution of average grain size with temperature and time can be expressed by Eqs. (8)-(9)[79–81].

$$D^2 - D_0^2 = kt \tag{8}$$

$$k = k_0 \exp\left(-\frac{Q}{RT}\right) \tag{9}$$

where D is the average grain size after solution treatment, D_0 is the initial grain size, k is the diffusion coefficient, k_0 is a constant related to the material, Q is the activation energy for diffusion, R is the gas

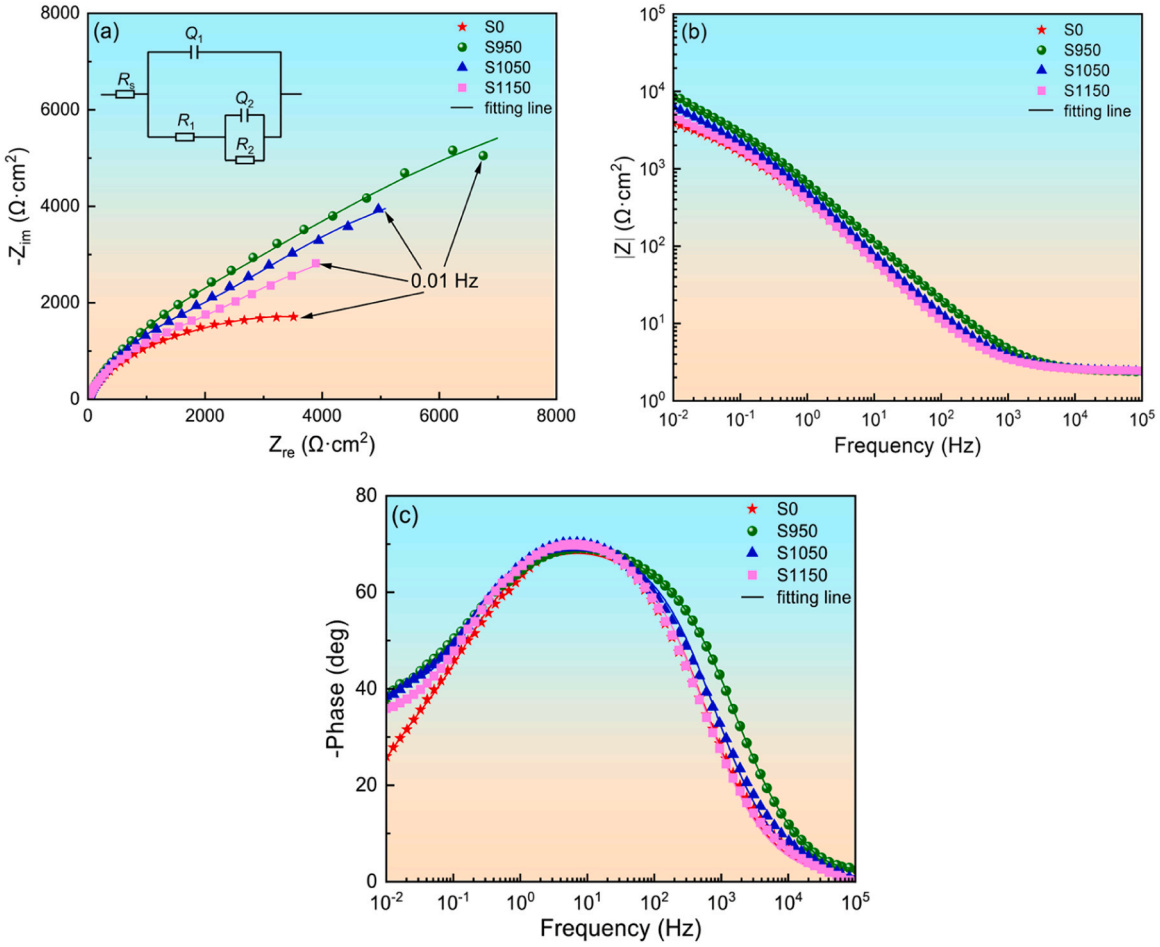


Fig. 11. EIS curves of the LDSs after potentiostatic polarization in 3.5 wt% NaCl solution: (a) Nyquist plot, (b) Bode magnitude plot, (c) Bode phase plot.

Table 3
Fitting parameters of the EIS of the LDSs after potentiostatic polarization.

	R_s (Ω·cm^2)	Q_1 (Ω^{-1}·cm^{-2}s^n)	n_1	R_1 (Ω·cm^2)	Q_2 (Ω^{-1}·cm^{-2}s^n)	n_2	R_2 (Ω·cm^2)	R_p (Ω·cm^2)
S0	6.50	6.50×10^{-4}	0.79	2736	1.79×10^{-3}	0.70	3407	6143
S950	6.34	3.58×10^{-4}	0.77	6345	7.94×10^{-4}	0.74	15010	21355
S1050	6.45	4.82×10^{-4}	0.79	4703	1.53×10^{-3}	0.83	9441	14144
S1150	6.52	5.85×10^{-4}	0.80	3497	2.07×10^{-3}	0.78	7259	10756

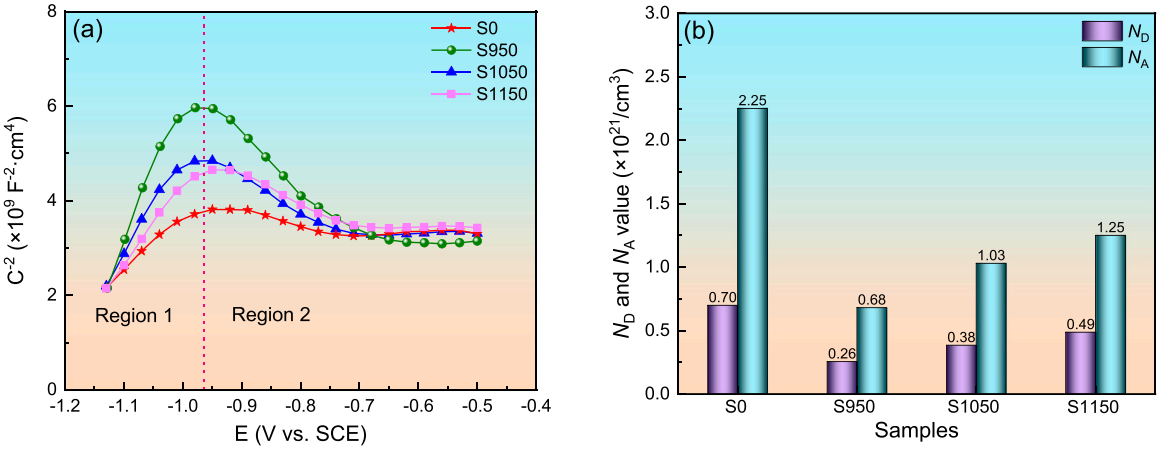


Fig. 12. (a) Mott-Schottky curves and (b) The N_D and N_A curves of LDSs in 3.5 wt% NaCl solution.

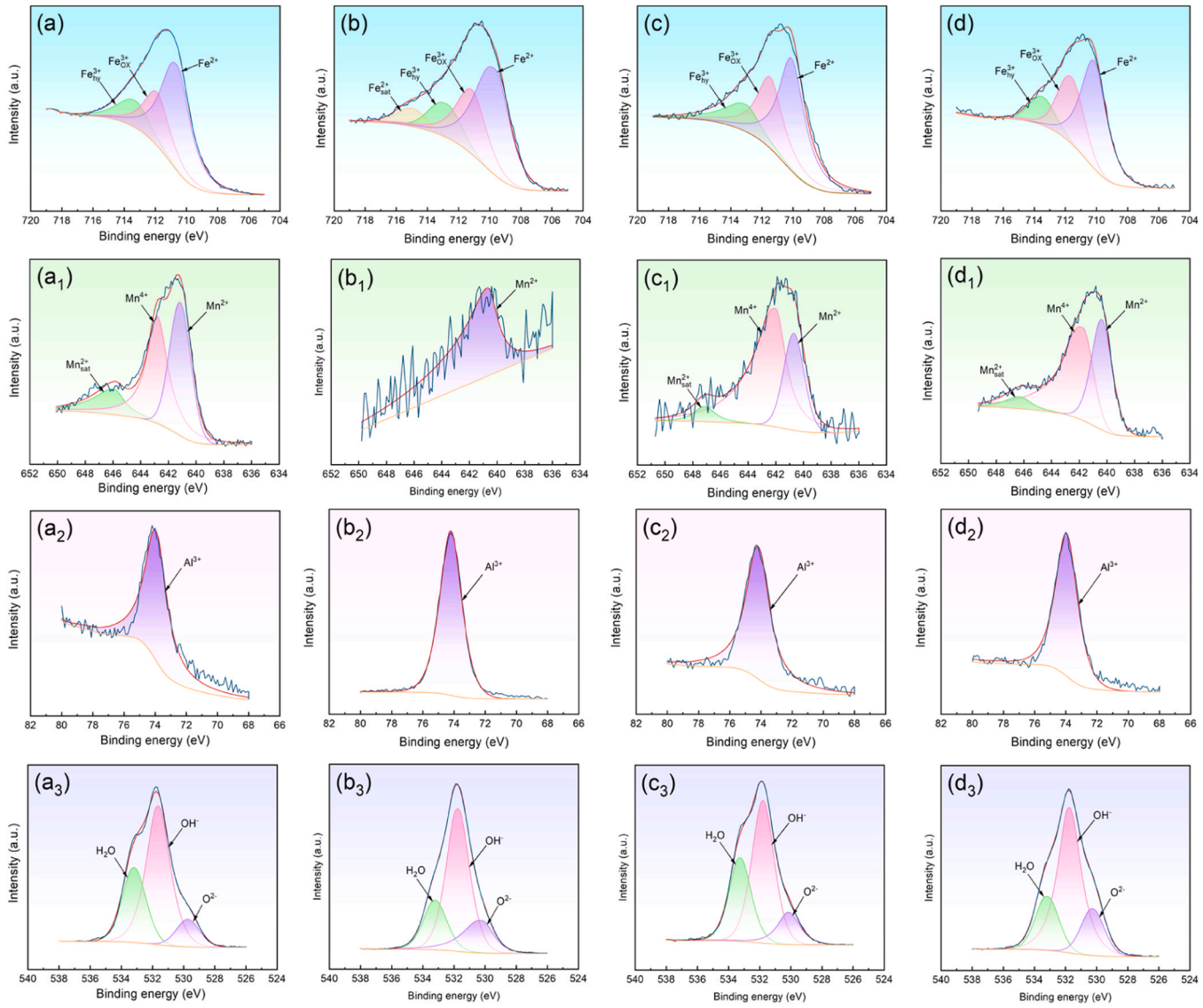


Fig. 13. XPS spectra of Fe2p$_{3/2}$, Mn2p$_{3/2}$, Al2p, and O1s in the oxide film of the LDSs: (a-a$_3$) S0, (b-b$_3$) S950, (c-c$_3$) S1050, (d-d$_3$) S1150.

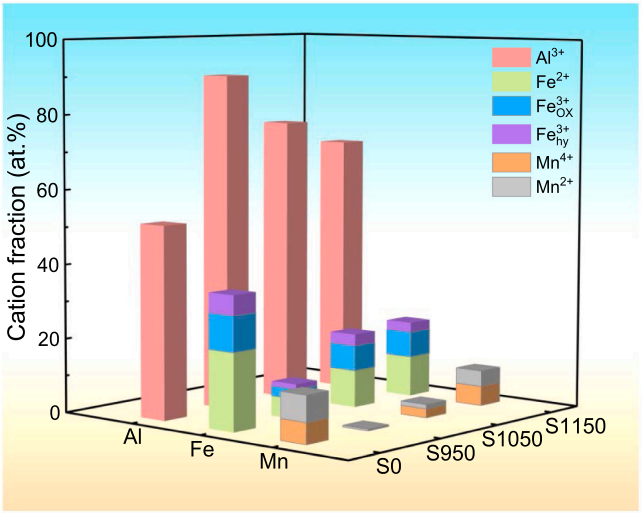


Fig. 14. Relative content of different valence states of metal elements in the oxide film of the LDSs.

constant, and T is the Kelvin temperature.

With increasing solution temperature, the diffusion coefficient k increases exponentially, which enhances the atomic mobility at elevated temperatures. This facilitates the migration of GBs, where large grains progressively swallow small grains, thereby accelerating grain growth. Additionally, GB energy plays a critical role in driving this process, particularly in the presence of HAGBs, which possess higher energy. The material always tends to reduce free energy[82,83], and GB migration is more pronounced at high temperatures. This process reduces the GB density and the total free energy of the system. Consequently, grain growth accelerates at high temperature. Conversely, ATBs represent low-energy special boundaries, which are gradually swallowed due to accelerated GB migration at higher temperatures. In summary, the solution temperature affects the grain size and GB density of LDSs through the synergistic effects of atomic diffusion and interfacial energy on GB migration behavior.

4.2. *Thermodynamic analysis of oxide film formation on the LDSs*

The oxide film on LDSs plays a crucial role in determining their corrosion resistance. The experimental results reveal that with increasing solution temperature, the Al content in the film decreases,

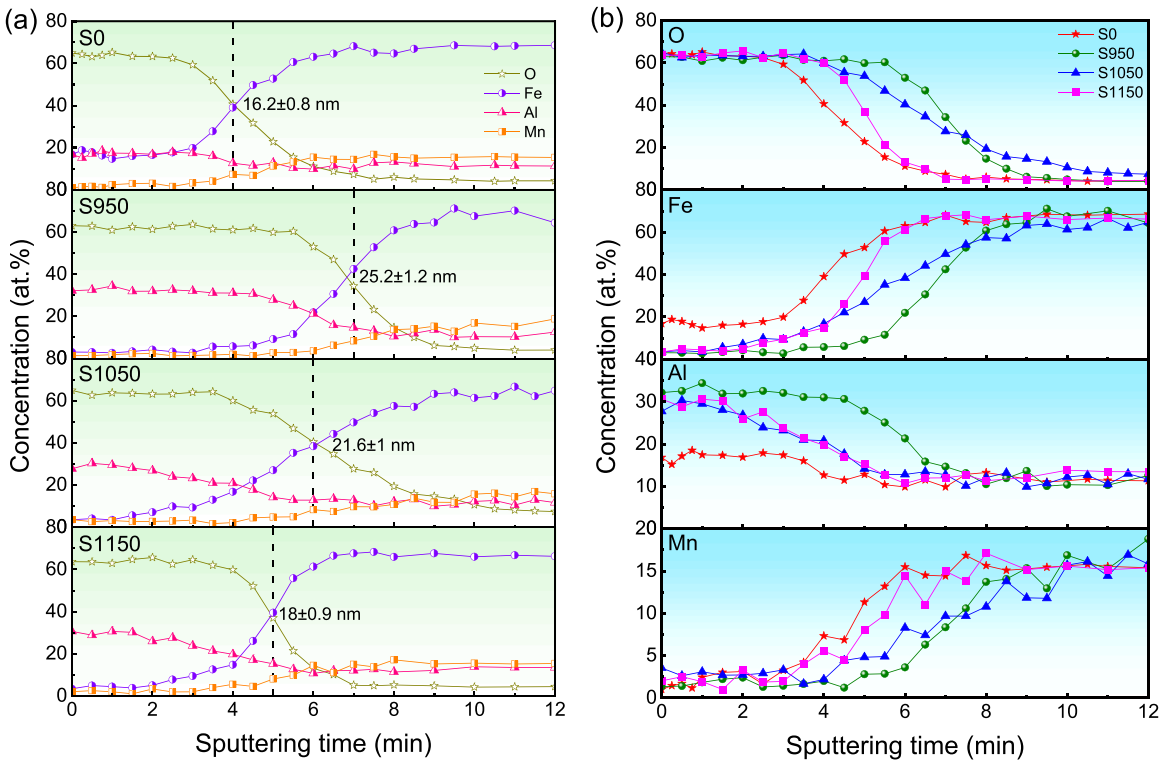


Fig. 15. AES images of the oxide film of the LDSs: (a) comparison between different samples, (b) comparison between different elements.

whereas the Fe and Mn contents increase, accompanied by a corresponding reduction in film thickness. This behavior can be explained by the strong oxidation tendency of Al and its selective dissolution behavior during the initial stage of corrosion. At room temperature, the standard Gibbs free energy of formation (ΔG°) values for the oxidation reactions of Al, Fe, and Mn are presented in Eqs. (10)-(12). The Ellingham diagram demonstrates that metals with lower ΔG° values tend to oxidize more strongly. Notably, the oxidation reaction curve of Al consistently

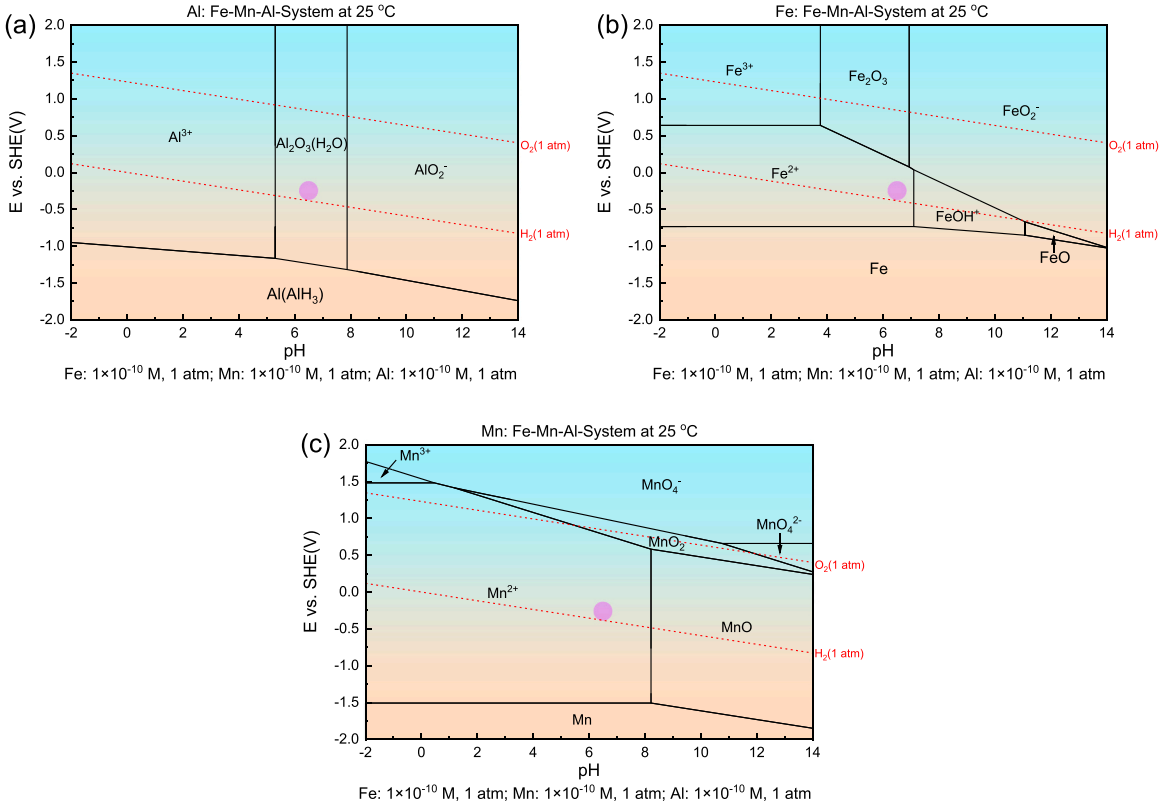


Fig. 16. E-pH diagram of the LDSs in Fe-Mn-Al-H_2O system: (a) Al, (b) Fe, (c) Mn, and the pink circles represent the experimental conditions of this study.

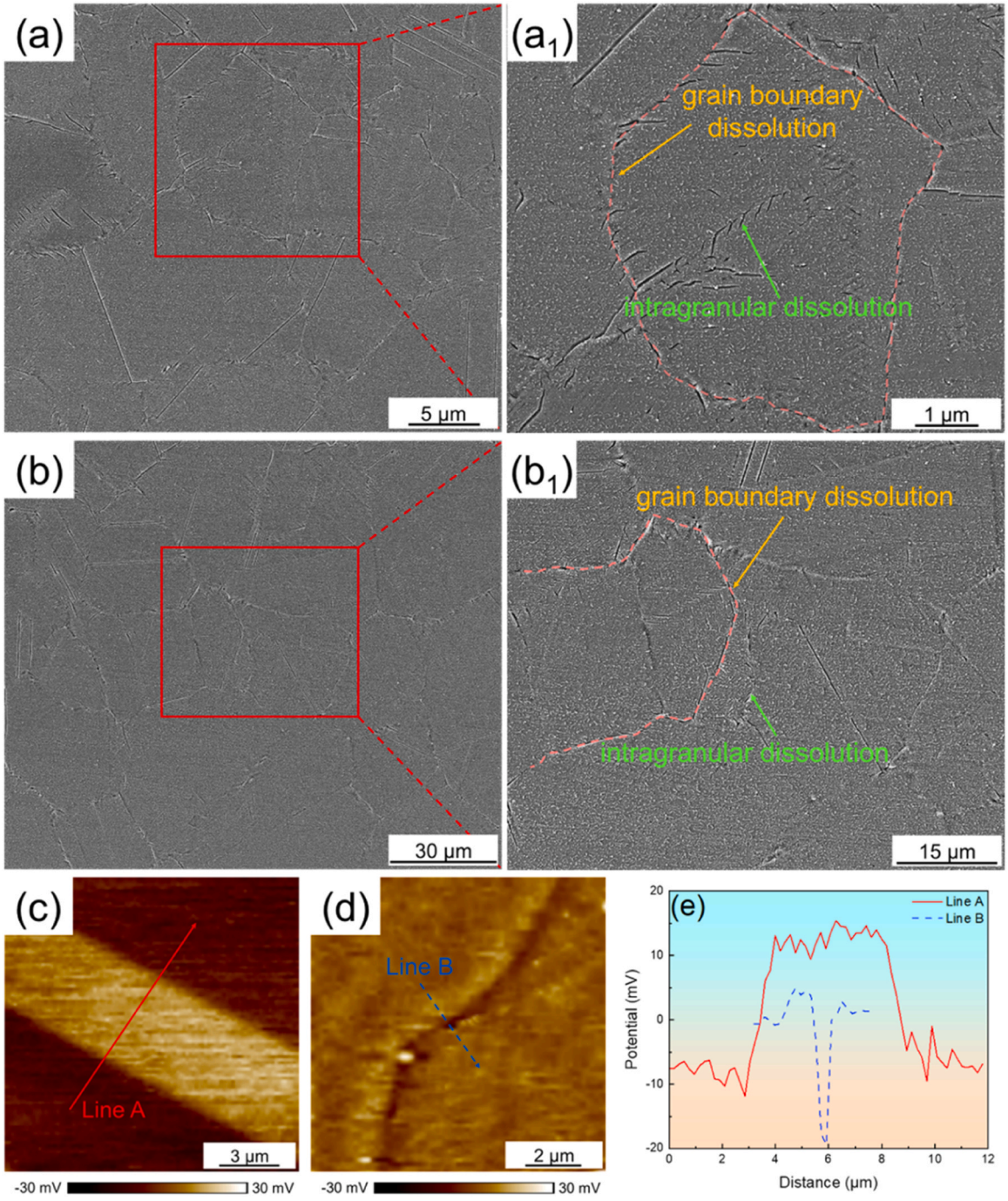


Fig. 17. Dissolution behavior of the LDSs with different grain size at the initial stage of corrosion: (a-a_1) S950, (b-b_1) S1150 and surface potential distribution: (c) ATBs, (d) HAGBs, (e) potential distribution.

appears at the bottom of the Ellingham diagram. This shows that Al has the strongest thermodynamic driving force for oxidation in all the temperature ranges. Even in the deoxygenated environment, Al preferentially oxidizes to form thermodynamically stable Al_2O_3. In contrast, the oxidation reactions of Fe and Mn have higher ΔG° values, making them less prone to oxidation in the deoxygenated environment. From an electrochemical perspective, the standard electrode potentials of these three elements are shown in Eqs. (13)-(15). More negative standard electrode potentials correlate with greater metal oxidation susceptibility. Al exhibits the most negative potential, confirming its highest propensity for electron loss and anodic dissolution, which provides the electrochemical driving force for its oxidation process.

$$4Al + 3O_2 \rightarrow 2Al_2O_3 \ \Delta G^\circ = -756.3 \text{ kJ/mol} \quad (10)$$

$$2Fe + O_2 \rightarrow 2FeO \ \Delta G^\circ = -116.7 \text{ kJ/mol} \quad (11)$$

$$2Mn + O_2 \rightarrow 2MnO \ \Delta G^\circ = -173.4 \text{ kJ/mol} \quad (12)$$

$$Al \rightarrow Al^{3+} + 3e^- \ E^\circ = -1.67 \text{ V}_{SHE} \quad (13)$$

$$Mn \rightarrow Mn^{2+} + 2e^- \ E^\circ = -1.05 \text{ V}_{SHE} \quad (14)$$

$$Fe \rightarrow Fe^{2+} + 2e^- \ E^\circ = -0.44 \text{ V}_{SHE} \quad (15)$$

The results clearly demonstrate the pronounced propensity of Al for oxide film formation from both electrochemical and thermodynamic

perspectives. Al readily loses electrons to form Al^{3+}, which rapidly oxidizes and hydrolyses to produce Al_2O_3, resulting in the development of an Al-rich surface oxide layer. This protective layer effectively inhibits the oxidation of Fe and Mn, as illustrated in Eq. (16). At film defect sites, limited quantities of Fe and Mn react as shown in Eqs. (17)-(21) to form corresponding oxides. Although the reaction of Fe^{3+} hydrolysis to form Fe_2O_3 and FeOOH is thermodynamically stable and theoretically capable of oxide film formation[84], the initial anodic dissolution of Fe involves a two-step electrochemical process ($Fe \rightarrow Fe^{2+} \rightarrow Fe^{3+}$) with a high overall energy barrier. Moreover, the formation of Fe^{3+} is more dependent on the solution oxygen content, which is limited in deoxygenation systems. Consequently, Fe predominantly exists as Fe^{2+} in the oxide film, with only minor fractions further oxidized to Fe_2O_3 and FeOOH, but its film formation rate and film coverage are lower compared to Al. The dissolution behavior of Mn is analogous to that of Fe, but the driving force required for the formation of its oxidation products is weaker and limited by the dissolved oxygen concentration. As a result, Mn has the lowest enrichment level in the oxide film, with only trace amounts of $Mn(OH)_2$ or MnO_2 deposited within the film.

$$2Al^{3+} + 3H_2O \rightarrow Al_2O_3 + 6H^+ \quad (16)$$

$$Fe^{2+} + H_2O \rightarrow FeO + 2H^+ \quad (17)$$

$$2Fe^{3+} + 3H_2O \rightarrow Fe_2O_3 + 6H^+ \quad (18)$$

$$Fe^{3+} + 2H_2O \rightarrow FeOOH + 3H^+ \quad (19)$$

$$Mn^{2+} + 2H_2O \rightarrow Mn(OH)_2 + 2H^+ \quad (20)$$

$$Mn^{4+} + 2H_2O \rightarrow MnO_2 + 4H^+ \quad (21)$$

To comprehensively evaluate the stability of the Fe, Mn, and Al oxidation products under corrosive conditions, the E-pH diagram of the Fe-Mn-Al-H_2O system at 25 ℃ was calculated via HSC Chemistry 9 software. The calculation results are presented in Fig. 16. The results reveal that Al resides within the stable zone of Al_2O_3 under environmental conditions, which confirms its ability to spontaneously form a stable and dense oxide film even under anaerobic conditions. The main stable state of Fe is Fe^{2+}, although the thermodynamic stability zone of FeO in the E-pH diagram (Fig. 16b) is located in the high pH and negative potential region. In the current environment, Fe^{2+} is difficult to oxidize further because of the low oxygen concentration. Therefore, FeO remains the main oxide of Fe, with minor quantities of Fe_2O_3 and FeOOH potentially forming at the film defects or localized high-potential regions. The E-pH diagram of Mn (Fig. 16c) demonstrates that the minimal thermodynamic driving force for oxide formation under the current environmental conditions primarily exists as Mn^{2+}. The stable regions of Mn^{2+} and MnO_2 exhibit close proximity and comparable ranges, which suggests the thermodynamic feasibility of Mn oxidation to higher valence states within specific potential windows.

4.3. *Effects of grain size on oxide film formation and corrosion resistance of the LDSs*

The corrosion resistance of the LDSs exhibits a progressive decline with increasing grain size and decreasing GB density, with the S950 sample displaying superior corrosion resistance. This variation in corrosion resistance stems principally from the characteristics of the oxide film on the LDSs, which are predominantly governed by the chemical composition and microstructure of the steel[85–87]. Nevertheless, experimental evidence and analytical results indicate that grain size and GB density constitute the primary distinguishing features among current LDSs. Therefore, the difference in corrosion behavior can be reasonably attributed to changes in the oxide film properties resulting from grain size variation.

Previous studies[88–92] reported that GB segregation of alloying elements such as Cr and V significantly influences the corrosion resistance of the material. These studies suggest that elemental segregation at GBs may either promote the formation of stable passive films or lead to precipitates that serve as pitting initiation sites, thereby affecting corrosion performance. In contrast, the present study reveals a relatively homogeneous elemental distribution without noticeable GB segregation. The oxide film formation mechanism in this case appears to be predominantly governed by the intrinsic structural characteristics of GBs, which act as inherent surface defects of the crystal. Specifically, the crystallographic misorientation between adjacent grains causes substantial atomic disorder at HAGBs, which generates structural defects that include lattice distortion, vacancy accumulation, and coordination deficiencies[93,94]. These defects thermodynamically increase the interfacial free energy at GBs to levels substantially higher than those in grain interiors, thereby enhancing their susceptibility to anodic dissolution in electrochemical systems[95–98]. From a kinetic perspective, the weaker binding energy of boundary atoms decreases the activation energy required for metal ion dissolution from the crystal lattice. The structural disorder coupled with vacancy accumulation creates short-circuit diffusion pathways, which significantly promote the dissolution activity of GBs in corrosive environments[99–102]. When selective anodic dissolution initiates at GB, the local metal ion concentration rises sharply while the interfacial pH value declines. This acidification effect narrows the thermodynamic passive region and lowers the critical energy barrier for oxide film nucleation[100,103]. Concurrently, GBs serve as continuous permeation networks that facilitate both the inward migration of water molecules and the outward diffusion of metal cations. The synergistic interaction of these processes within GB defect regions enables rapid nucleation and growth of oxide films[104,105]. Consequently, even without elemental segregation, GBs govern both the initial dissolution behavior and subsequent oxide film formation due to their higher interfacial free energy and lower dissolution barriers.

Fig. 17 illustrates the dissolution behavior and surface potential distribution of LDSs with varying grain size during the initial stage of corrosion. SEM images of the S950 and S1150 samples (Fig. 17a-b) demonstrate preferential grain boundary dissolution during the initial stage of corrosion, resulting in the formation of continuous shallow etching grooves, whereas the grain interiors exhibit comparatively less dissolution. The surface potential distribution results in Fig. 17c-e demonstrate that the ATBs exhibit a higher potential than the matrix, while the HAGBs show the lowest potential. This potential distribution indicates that the HAGBs have the most active electrochemical reactions during the initial stage of corrosion and undergo preferential anodic dissolution. This process creates favorable conditions for the rapid oxidation of reactive elements such as Al[106,107], promoting the formation of an Al-rich oxide film. Notably, localized dissolution occurs at certain ATBs, as shown in Fig. 17, which is attributable to the approximately 30 mV potential difference between ATBs and the matrix. Despite their inherent stability, this potential disparity facilitates microgalvanic couples at the interface of the ATBs and the matrix, which establishes anodic regions in the adjacent matrix and induces metal dissolution extending along the twin boundary. The uniform distribution of ATBs contributes to maintaining the structural continuity and composition homogeneity of the oxide film in corrosive environments and promotes uniform oxide film growth across the entire grain surface.

Conversely, larger grain sizes correspond to lower GB densities, resulting in a sparsely distributed GB region that diminishes the number of reaction sites for oxide film formation during the initial stage of corrosion. This reduction in active sites results in a slower film formation rate and produces a more discontinuous oxide layer. Additionally, an increasing grain size correlates with a decreased Al content within the oxide film, producing a thinner and compositionally heterogeneous film exhibiting higher density of defects (Mott-Schottky results, Fig. 12). These defective regions may become channels for the infiltration of corrosive media, potentially leading to localized film breakdown and subsequent corrosion initiation[32,108,109].

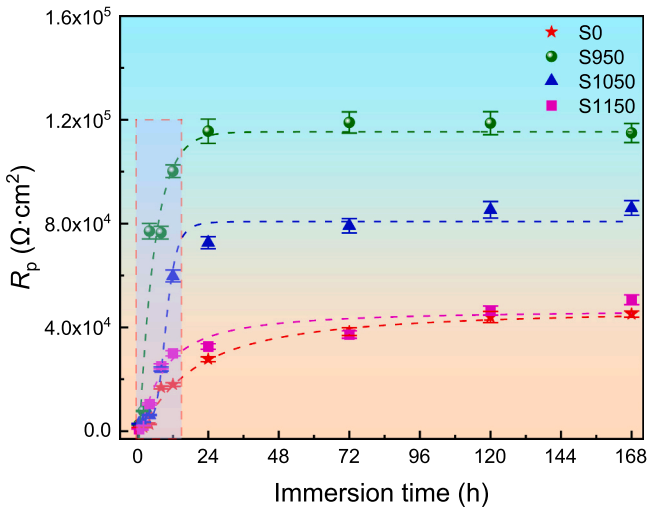


Fig. 18. The R_p values obtained from EIS fittings of the surface oxide film on the LDSs after immersion for different times.

To systematically investigate the differences in the oxide film formation rates among the various samples, the samples were immersed for periods ranging from 0.5 to 168 h under identical testing conditions, and the R_p values were statistically analyzed via EIS measurements. The results are shown in Fig. 18. The fine-grained sample (S950) exhibits consistently higher R_p values throughout the entire immersion period, reaching 114.9 kΩ·cm^2 at 168 h, which is significantly higher than that of the coarse-grained sample (S1150) at 50.6 kΩ·cm^2. The R_p value of the S950 sample increases rapidly during the initial stage of immersion (2–8 h) and then stabilizes after approximately 24 h. This indicates that a dense and stable oxide film forms rapidly on the surface of the S950 sample, which effectively protects the penetration of the corrosive medium. Conversely, the coarse-grained samples display lower initial R_p values and slower growth rates with the extension of immersion time, suggesting delayed film formation and compromised structural integrity, which diminishes the protective efficacy. The variation among parallel samples is within 5 %, and the impedance evolution trend is consistent. The growth process of the oxide film has good repeatability and reproducibility in repeated experiments.

The Al content in the surface oxide films of the different LDSs significantly differed, as shown in Figs. 14 and 15. Notably, the S0 sample without solution treatment retains a substantial amount of κ-carbide, which limits the solid solubility of Al and yields the lowest Al content in its oxide film. Fig. 19a-b present the cyclic voltammetry curves of the LDSs in 3.5 wt% NaCl solution. The peak observed in the range of −0.81 V_{SCE}~-0.77 V_{SCE} corresponds to Al oxidation, whereas the oxidation peak near approximately −0.56 V_{SCE} represents Fe oxidation, possibly involving concurrent minor Mn oxidation.

In the first cycle (Fig. 19a), a distinct Al oxidation peak appears at approximately −0.81 V_{SCE}, with the S950 sample exhibiting the highest peak current density. This observation suggests the superior oxidation activity of surface Al in the S950 sample, which facilitates its rapid participation in initial oxide film formation. Comparatively, the Al oxidation peaks of the S1050 and S1150 samples are noticeably weaker, which demonstrates lower Al oxidation reactivity on their surfaces. Furthermore, quantitative analysis of the Al oxidation peak region during the first cycle reveals that the charge density of the S950 sample reaches 2.55 × 10^{-5} C/cm^2, which is higher than that of the S1050 (1.80 ×10^{-5} C/cm^2) and S1150 (1.66 ×10^{-5} C/cm^2) samples. This indicates that the oxidation reaction of Al in the S950 is more complete in the initial stage, which promotes the rapid formation of an Al-rich oxide film. Notably, the characteristic Fe oxidation peak near −0.56 V_{SCE} is absent in the first cycle. This result has been consistently confirmed through repeated experiments (see Supporting Information Fig. S1). This phenomenon arises because preferential oxidation and rapid film formation of Al coat the Fe surface with an Al-rich oxide layer, thereby restricting reaction sites and suppressing Fe oxidation, which prevents formation of a detectable current peak during the first cycle.

In the second cycle (Fig. 19b), a distinct Fe oxidation peak emerges at approximately −0.56 V_{SCE} alongside the persistent Al oxidation peak. The S950 sample displays the lowest peak intensity in this potential region. This result suggests the presence of a dense oxide film on the surface, which significantly inhibits the subsequent oxidation of Fe and Mn. Conversely, the S1050 and S1150 samples demonstrate substantially enhanced anodic peaks at this position, suggesting that the surface film is incompletely formed and that Fe and Mn continue to participate in the film formation process. The XPS results further confirm that the oxide film of the S950 sample contains a higher Al content coupled with reduced Fe and Mn concentrations compared to the other samples. This difference stems from the fine-grained structure and high GB density of the S950, which facilitate the preferential dissolution of Al at the GBs. Consequently, a continuous and dense Al-rich oxide film rapidly forms during the initial stages, effectively impinging subsequent Fe and Mn oxidation processes.

Fig. 20 shows the oxide film formation process and corrosion mechanism in LDSs with varying grain sizes. As shown in Fig. 20a-a$_3$, the fine-grained S950 sample has smaller grains and higher GB density, especially HAGBs and ATBs, facilitating the preferential dissolution of metallic elements (such as Al). This selective dissolution accelerates nucleation and growth of an Al-enrich oxide film, whereas the uniformly distributed ATBs promote homogeneous film formation, ultimately yields a denser, more continuous protective layer. The AES and Mott-

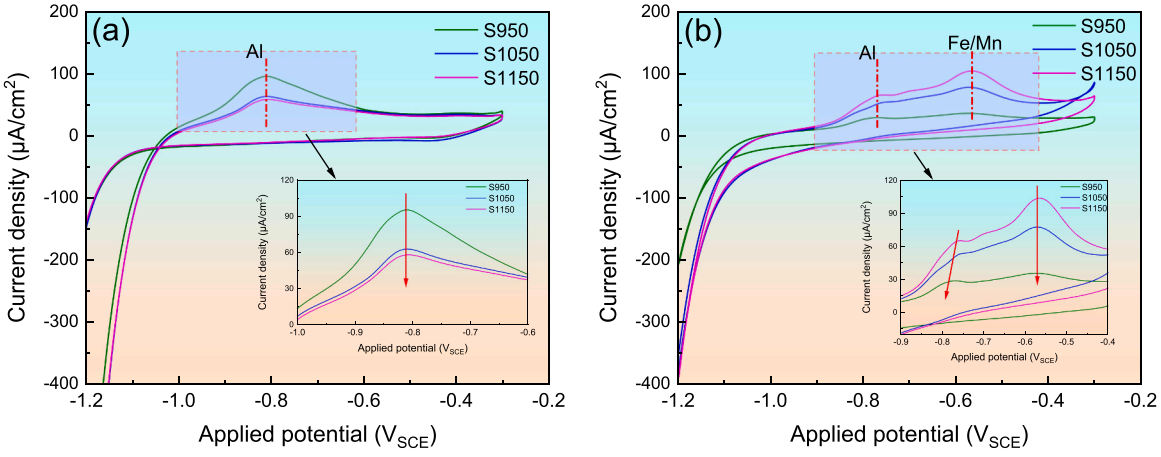


Fig. 19. Cyclic voltammetry curves of the LDSs in 3.5 wt% NaCl solution: (a) first cycle; (b) second cycle.

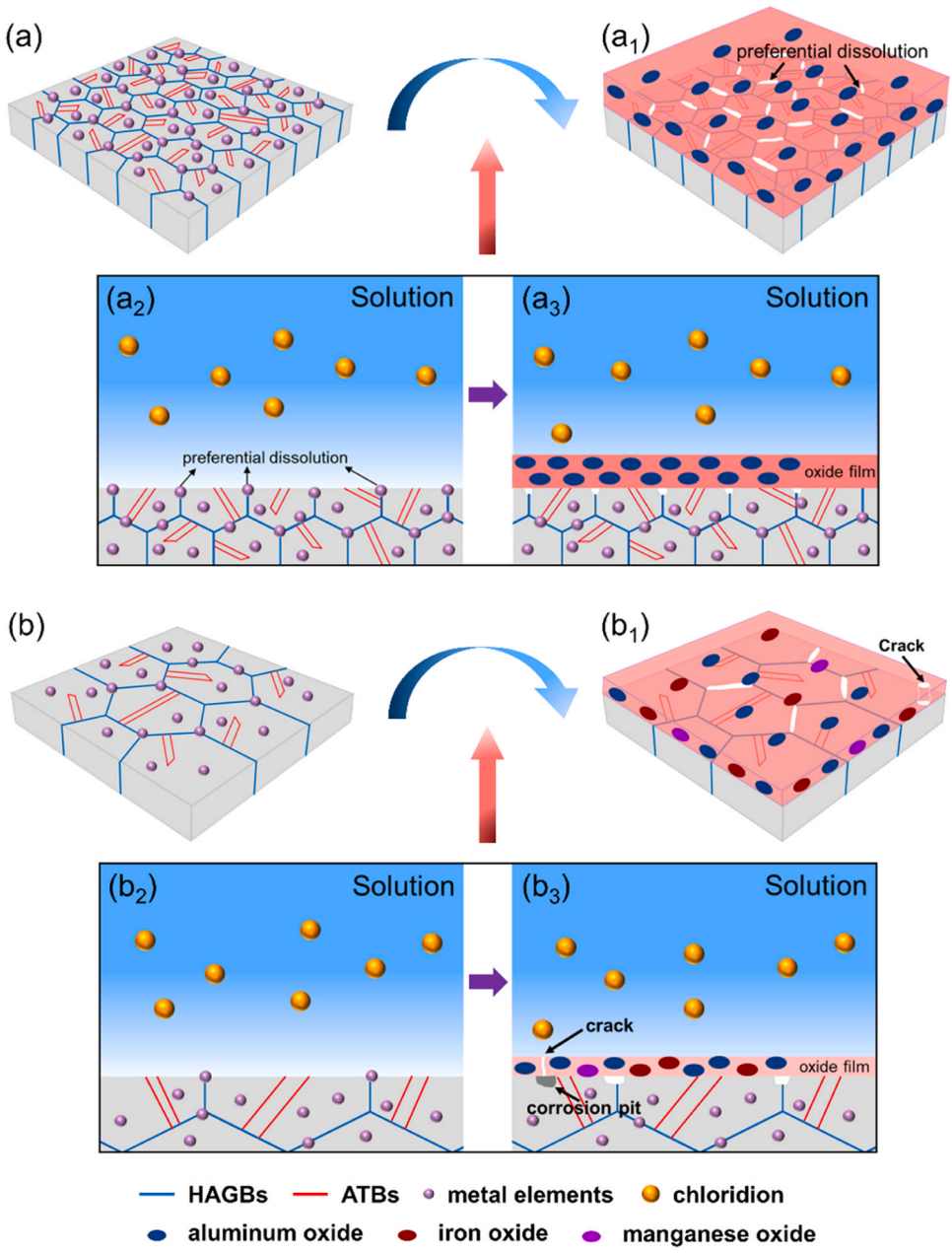


Fig. 20. Schematic diagrams of oxide film formation and corrosion mechanisms of the LDSs with different grain size: (a) small grain size sample; (b) large grain size sample.

Schottky analysis results further confirm that the oxide film on the S950 sample surface primarily consists of Al and exhibits a uniform compositional distribution and appropriate thickness. The film also contains a low density of defects, which effectively impedes the invasion of corrosive media and confers superior corrosion resistance. Conversely, the coarse-grained S1050 and S1150 samples have a reduced GB density and sparse boundary distribution, which retard initial oxide film formation and reduce uniform nucleation sites, as shown in Fig. 20b-b_3. Consequently, these samples develop thinner oxide films with diminished Al content and elevated Fe/Mn concentrations. This results in a structurally porous oxide layer prone to crack formation and other defects. These defective areas serve as infiltration channels for corrosive media, which reduces the protective efficacy of the oxide film, and ultimately degrades the corrosion resistance of LDSs.

5. Conclusions

This study investigates the influence of grain size variations induced by solution treatment on the microstructure and corrosion resistance of Fe-Mn-Al-C LDSs. The main conclusions are summarized as follows:

(1) Increasing the solution temperature promotes complete recrystallization in LDSs, and the microstructure contains abundant ATBs and HAGBs. The grain size increases progressively from 8.7 μm to 32 μm, while the HAGBs density decreases from 0.114 μm/μm^2 to 0.019 μm/μm^2.
(2) Electrochemical measurements indicate that the corrosion resistance of LDSs is negatively correlated with grain size. The fine-grained sample (S950) has the lowest i_{bre} (2.72 μA/cm^2) and the highest R_p (12486 Ω·cm^2). Its corrosion rate is 0.160 mm/y,

which is 43 % lower than that of the coarse-grained sample (S1150), and excellent corrosion resistance is exhibited.

(3) The oxide film of LDSs exhibits duplex semiconductor characteristics of n-type and p-type. Both the N_A and N_D values increase with increasing grain size. The N_A and N_D of the S950 sample are $0.68 \times 10^{21}/cm^3$ and $0.26 \times 10^{21}/cm^3$, respectively, which are significantly lower than those of the S1150 sample. This suggests that finer grains reduce defects within the oxide film. XPS analysis further reveals that grain coarsening reduces the Al content in the film from 90.17 at% to 69.91 at%, while increasing the Fe and Mn contents. Therefore, the stability of the film is reduced.

(4) This study reveals the mechanism of the effects of grain size on Al-rich oxide film formation and corrosion resistance in LDSs. These results indicate that metallic elements preferentially dissolve and form oxide films at the GBs. The change in grain size directly affects the formation rate and growth behavior of oxide films. The HAGBs provide numerous nucleation sites for the growth of the oxide film. A continuous and stable Al-rich oxide film forms quickly during the initial stage of corrosion, significantly enhancing the corrosion resistance. Conversely, coarse-grained microstructures impede oxide film growth, reduce the Al content, and degrade the film density and stability.

CRediT authorship contribution statement

Wenquan Cao: Validation, Supervision. **Yucheng Ji:** Writing – review & editing, Supervision, Methodology. **Xiaoqian Fu:** Validation, Investigation, Formal analysis. **Li Wang:** Writing – review & editing, Supervision, Methodology. **Guang Yang:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Chaofang Dong:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Jianxiong Liang:** Validation, Supervision. **Cunyu Wang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.corsci.2025.113331.

Data availability

No data was used for the research described in the article.

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